



US 20250282811A1

(19) **United States**(12) **Patent Application Publication****PADILLA IGLESIAS et al.**(10) **Pub. No.: US 2025/0282811 A1**(43) **Pub. Date: Sep. 11, 2025**

(54) **A 3-O-MONODESMOSIDIC SAPONINS EXTRACT, STABLE AT PHYSIOLOGICAL PH, METHOD FOR PREPARING IT FROM A QUILLAJA PLANT EXTRACT CULTURED WITH A MICROBIAL CONSORTIUM, USES AND SUCH MICROBIAL CONSORTIUM**

(71) Applicants: **DESERT KING CHILE S.A.**, Quilpué, Valparaíso (CL); **PONTIFICIA UNIVERSIDAD CATÓLICA DE VALPARAÍSO**, Valparaíso (CL)

(72) Inventors: **Leandro PADILLA IGLESIAS**, Quilpué, Valparaíso (CL); **Leda Marcela GUZMAN MALUENDA**, Curauma (CL)

(21) Appl. No.: **18/697,798**

(22) PCT Filed: **Oct. 2, 2021**

(86) PCT No.: **PCT/IB2021/059065**

§ 371 (c)(1),

(2) Date: **Apr. 2, 2024**

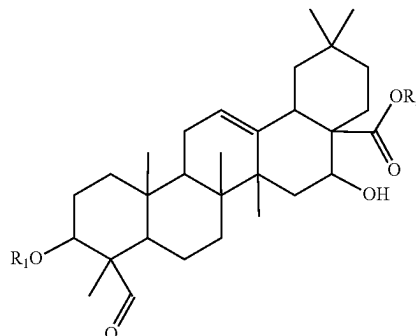
Publication Classification

(51) **Int. Cl.**
C07H 15/256 (2006.01)

(52) **U.S. Cl.**
CPC **C07H 15/256** (2013.01)

(57) **ABSTRACT**

The present invention relates to a biotransformation process effected by cultures of a microbial consortium for the preparation of a *Quillaja* plant extract with antibacterial and cytotoxic properties, comprising one or more monodesmosidic 3-O-glycosides of quillaic acid of formula (I),



wherein R_1 is {xylopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, {xylopyranosyl-(1→3)-glucopyranosiduronate salt}; {rhamnopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, {rhamnopyranosyl-(1→3)-glucopyranosiduronate salt}; or {galactopyranosyl-(1→2)-glucopyranosiduronic acid} or a salt thereof, {galactopyranosyl-(1→2)-glucopyranosiduronate salt}; or {glucopyranosiduronic acid} or a salt thereof, {glucopyranosiduronate salt}, and R_2 is hydrogen or a cation specie. Such microbial consortium comprising microorganisms selected from *Rhodotorula* sp having accession number RGM 3031 (Colony P) dated Jul. 22, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Microbacterium* sp having accession number RGM 3021 (Colony Y) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Mucor* sp having accession number RGM 2984 (Colony C) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Paenibacillus* sp and having accession number RGM 3032 (Colony W) dated Sep. 3, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM). The present invention is also related to the extract (SPH03) obtained from the above-described method, uses and such microbial consortium. The present extract is stable at physiological pH and has a strong reduced hemolytic effect, a cytotoxic effect on cancer cells and an antibacterial effect on oral pathogens such as *Streptococcus mutans*.

Specification includes a Sequence Listing.

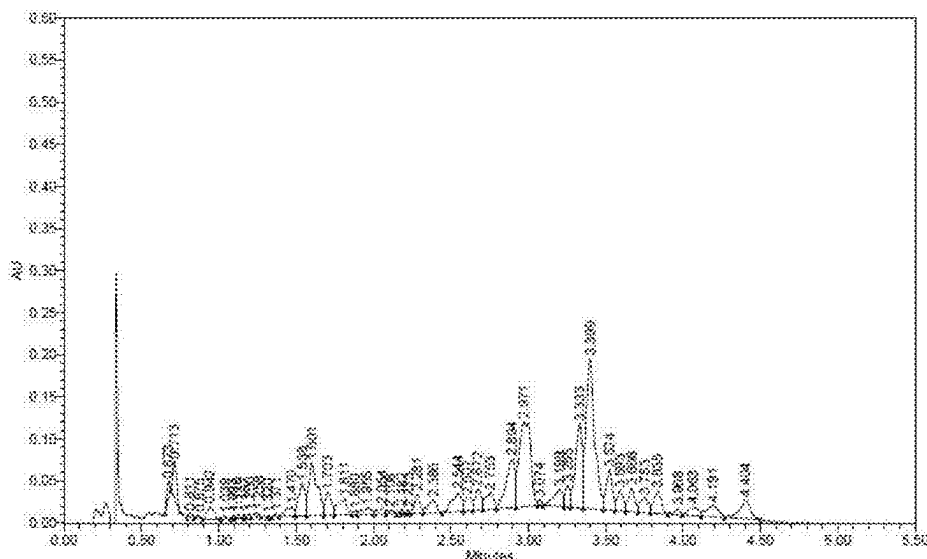


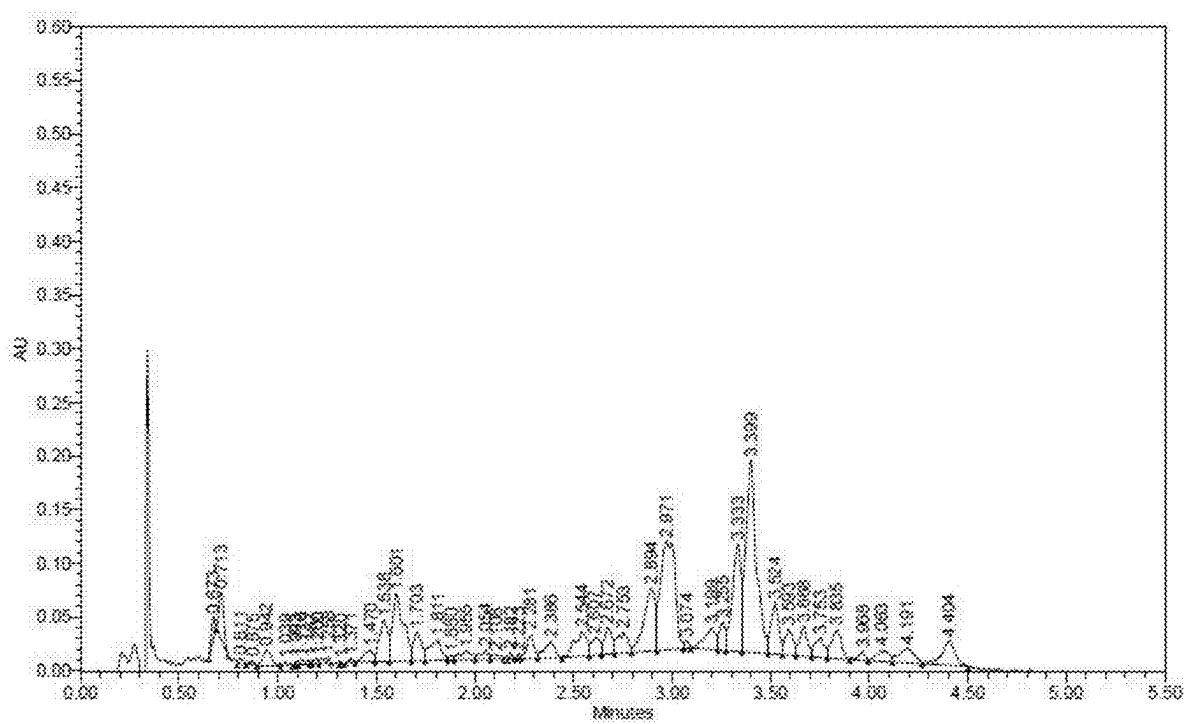
Figure 1

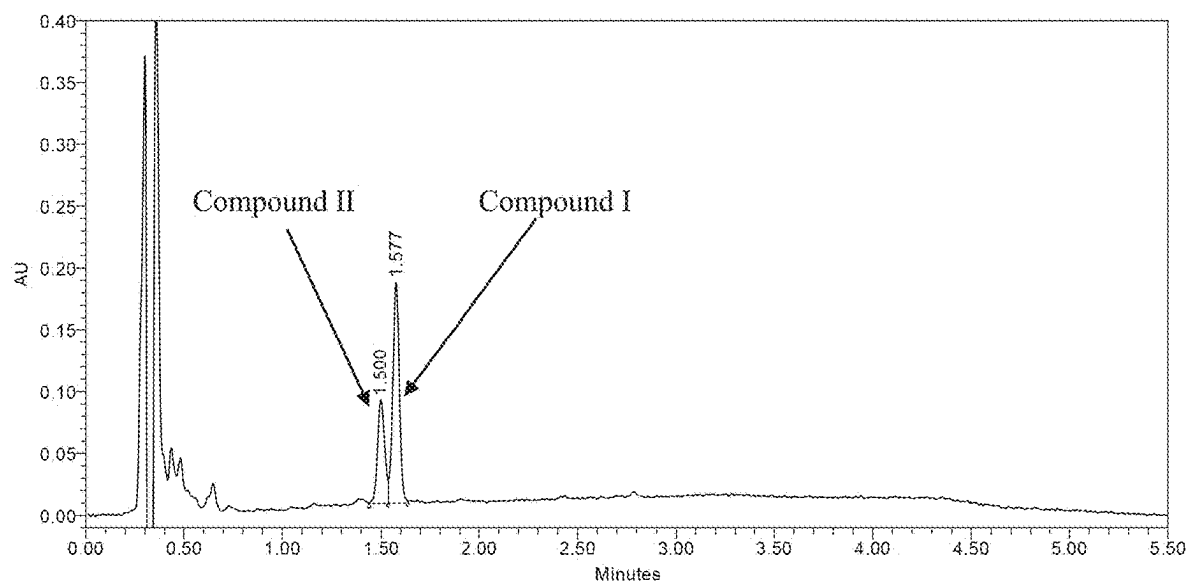
Figure 2

Figure 3

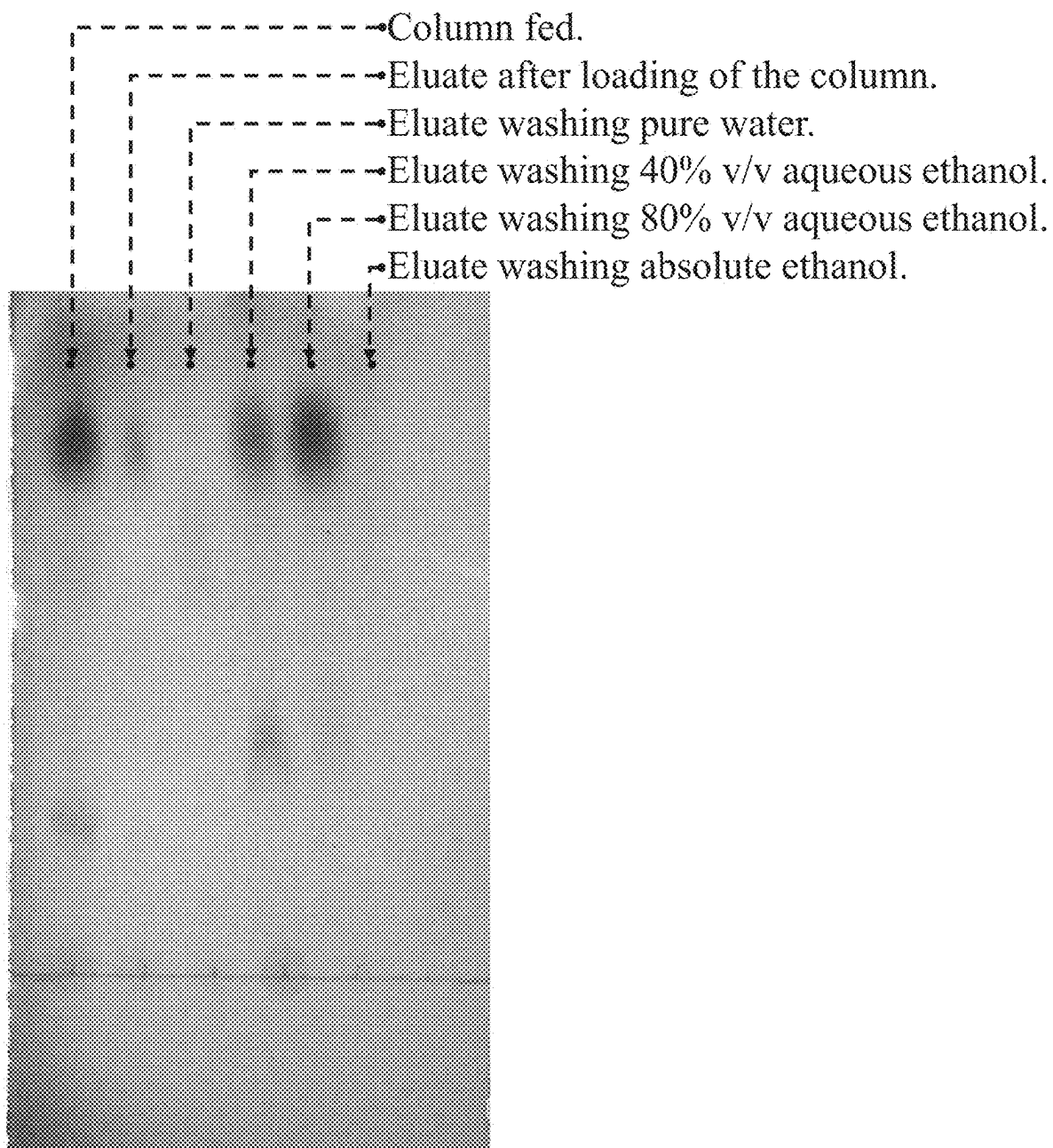


Figure 4

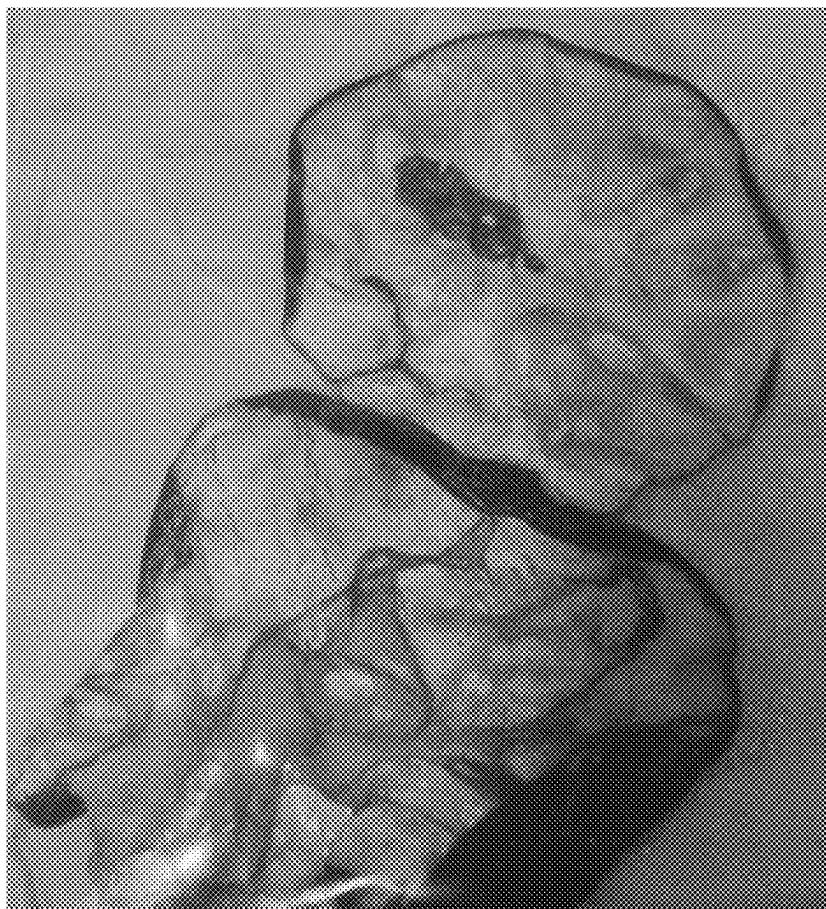


Figure 5

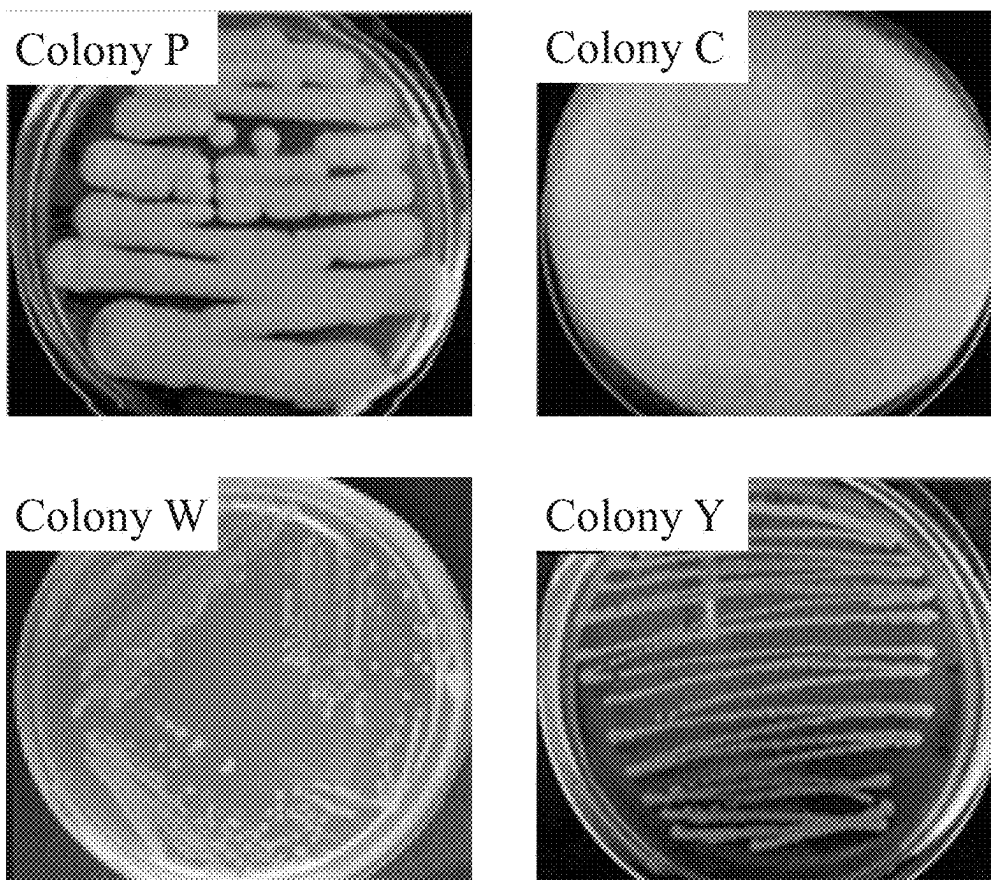


Figure 6A

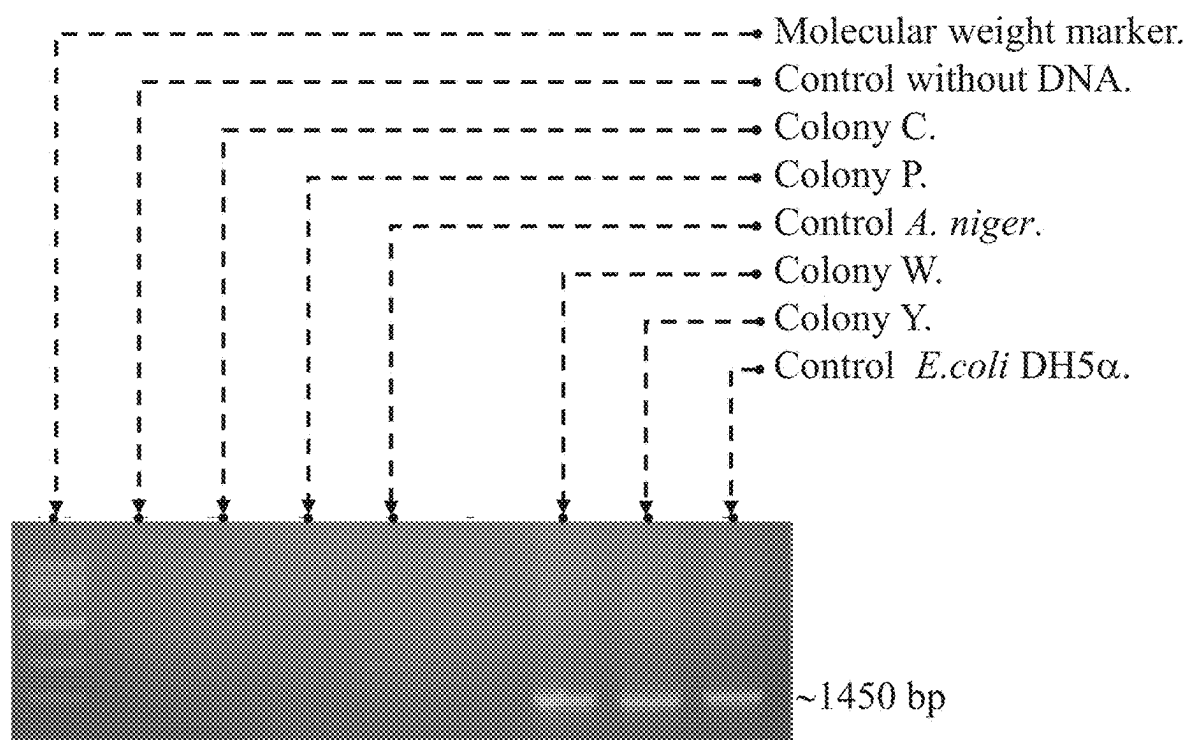


Figure 6B

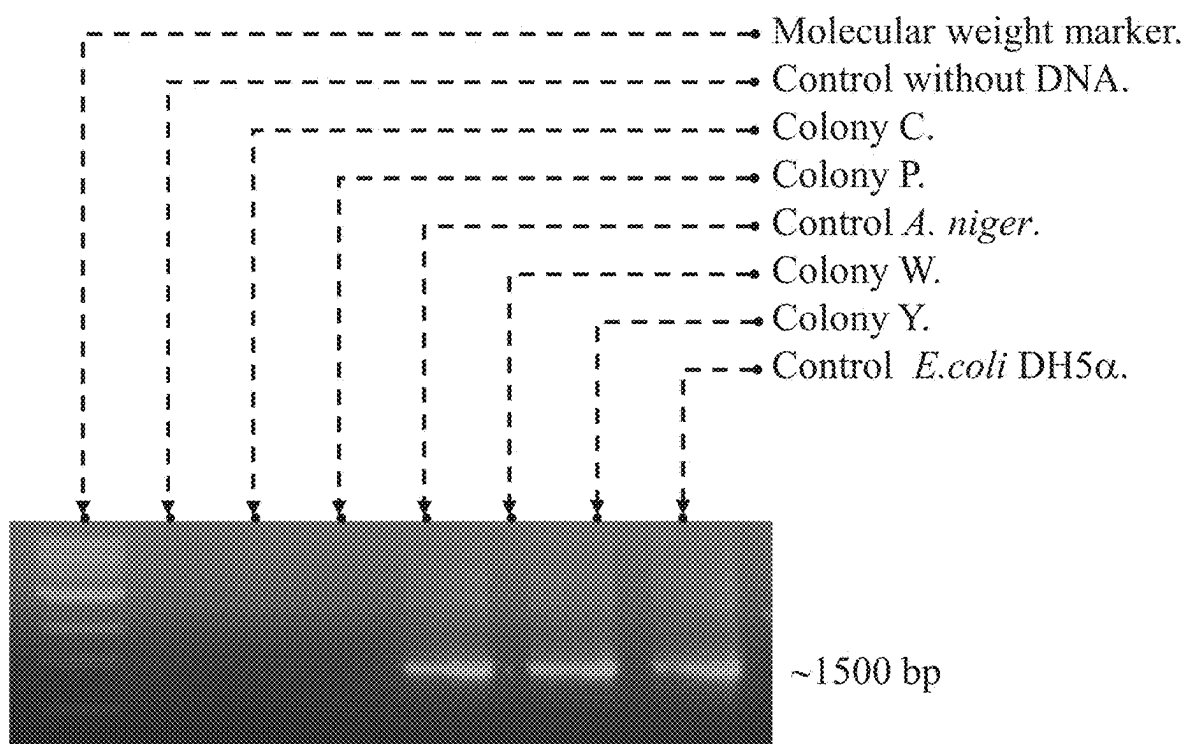


Figure 6C

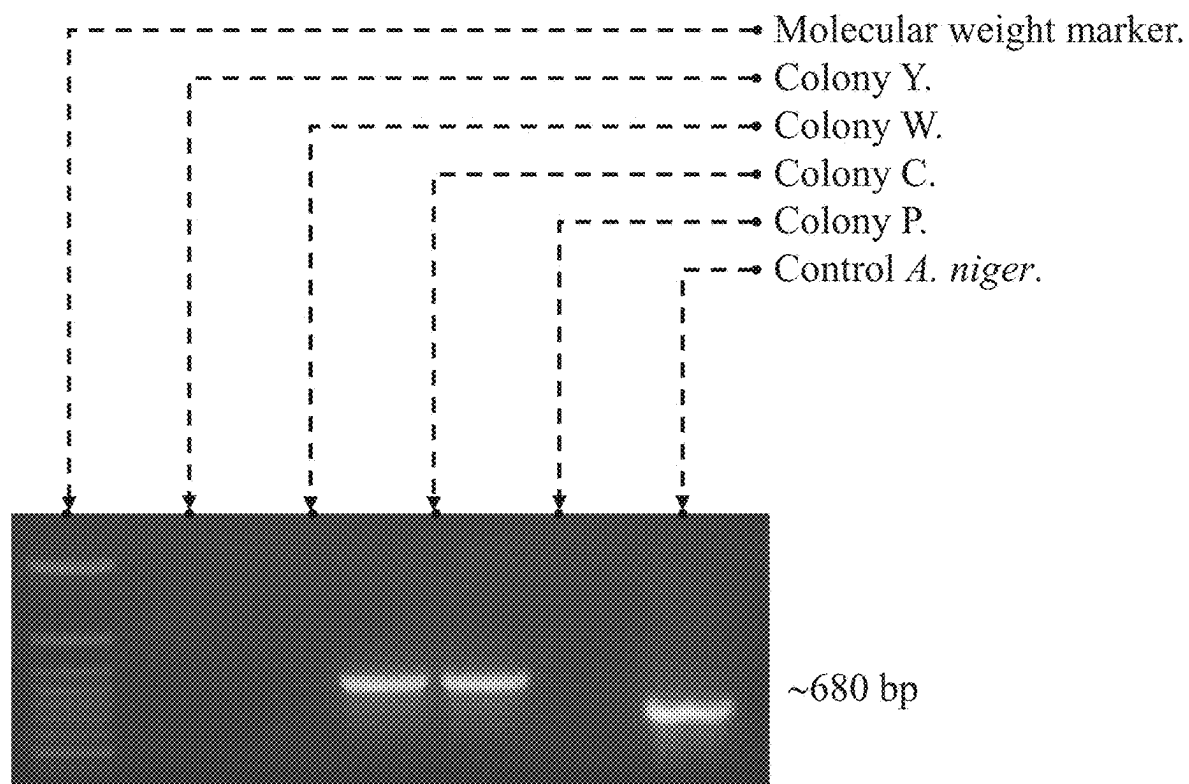


Figure 6D

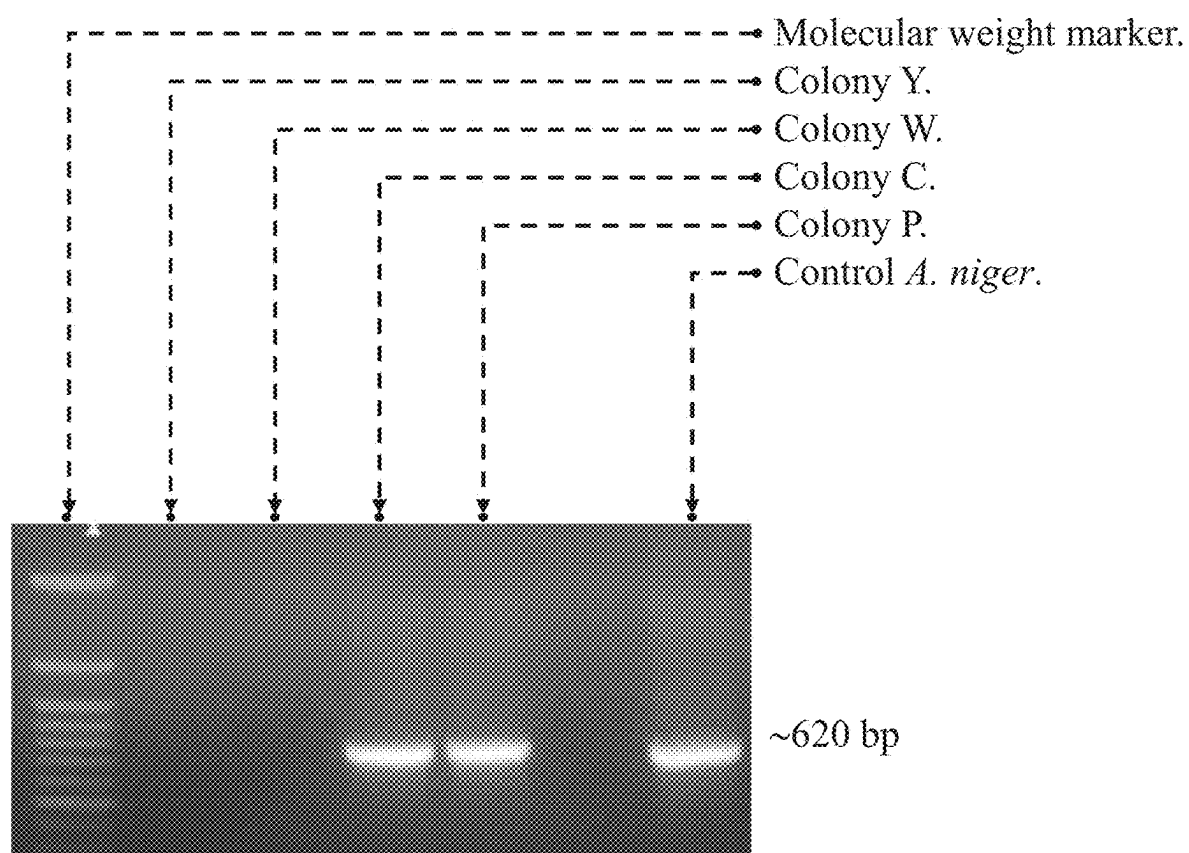


Figure 7

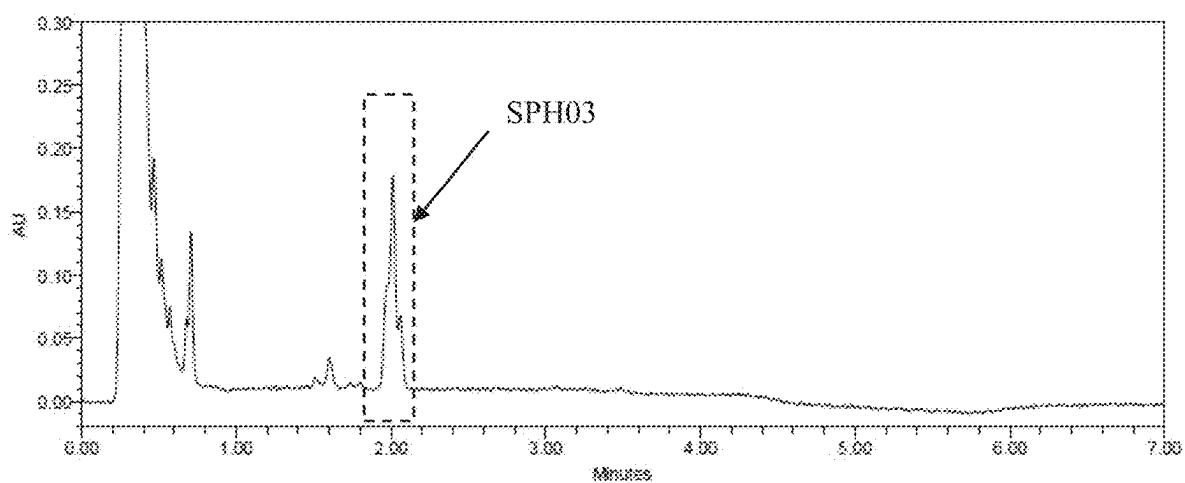


Figure 8

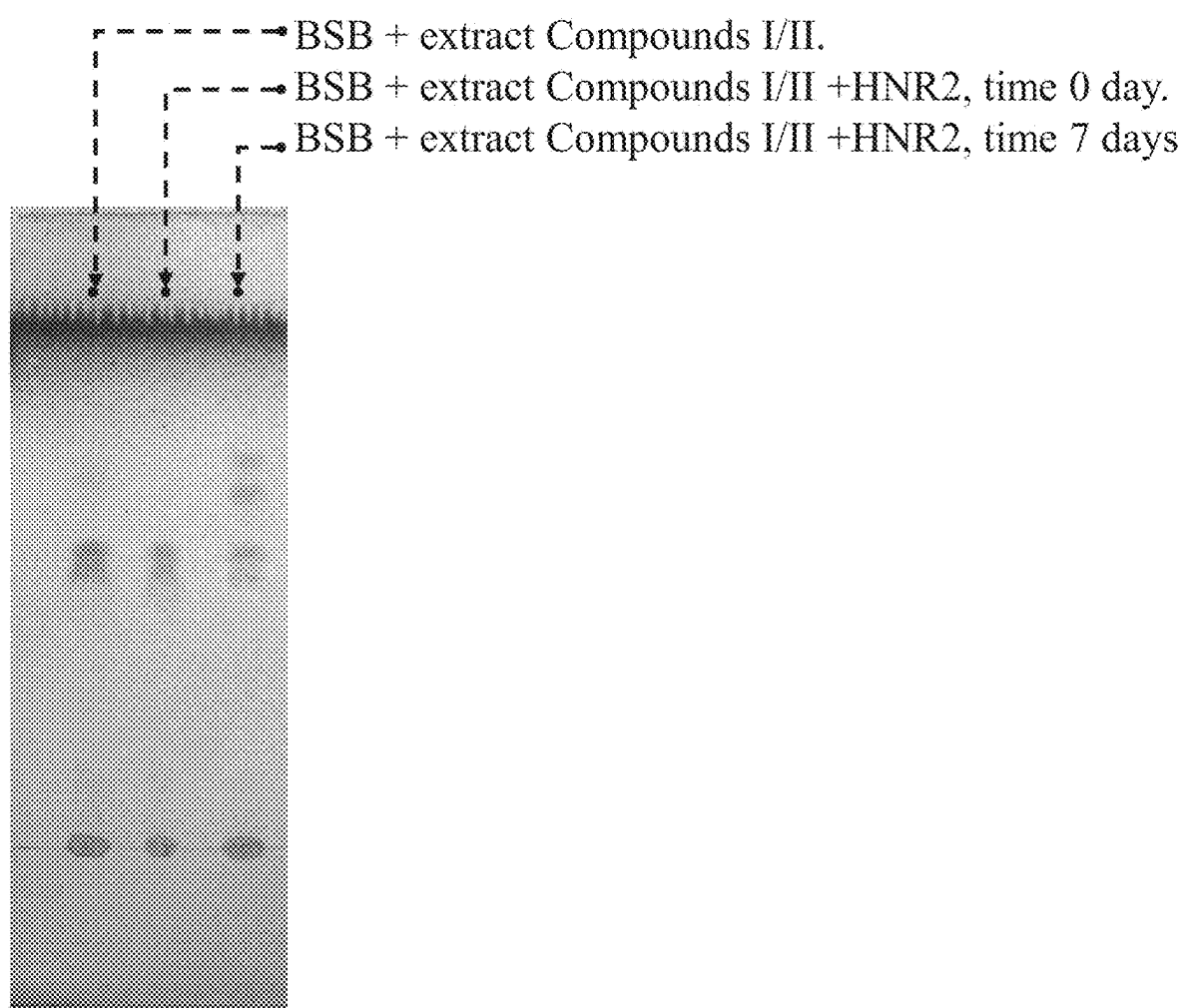


Figure 9

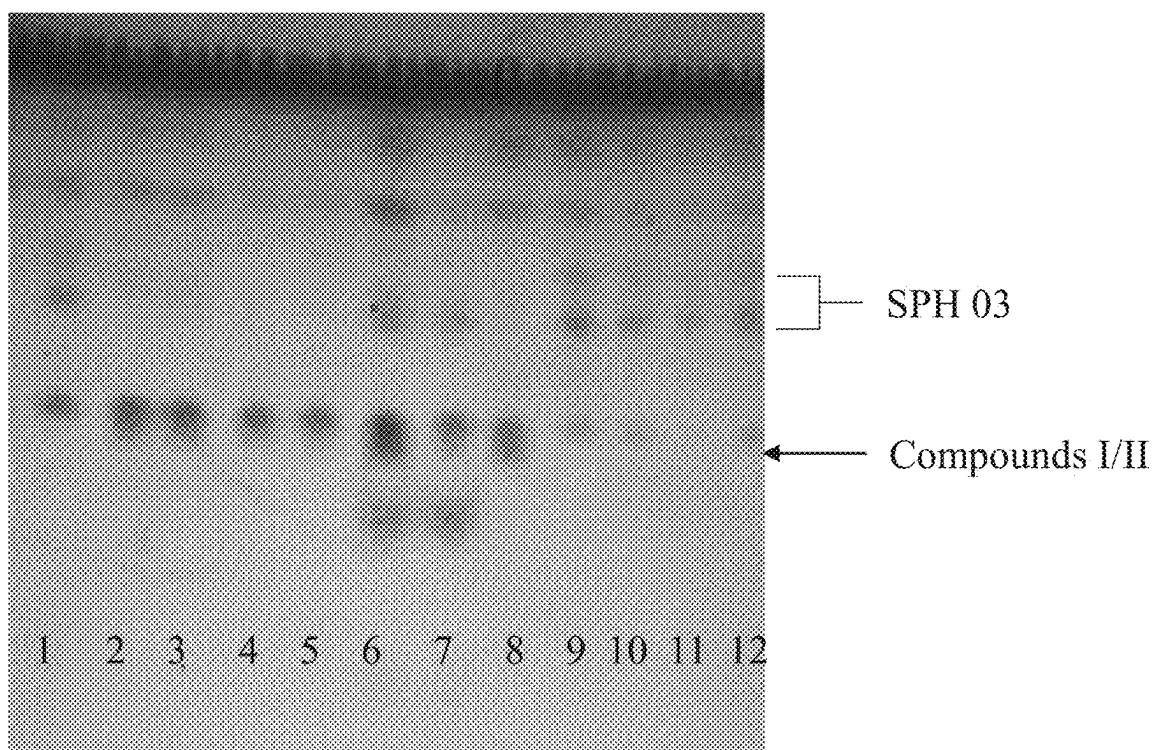


Figure 10

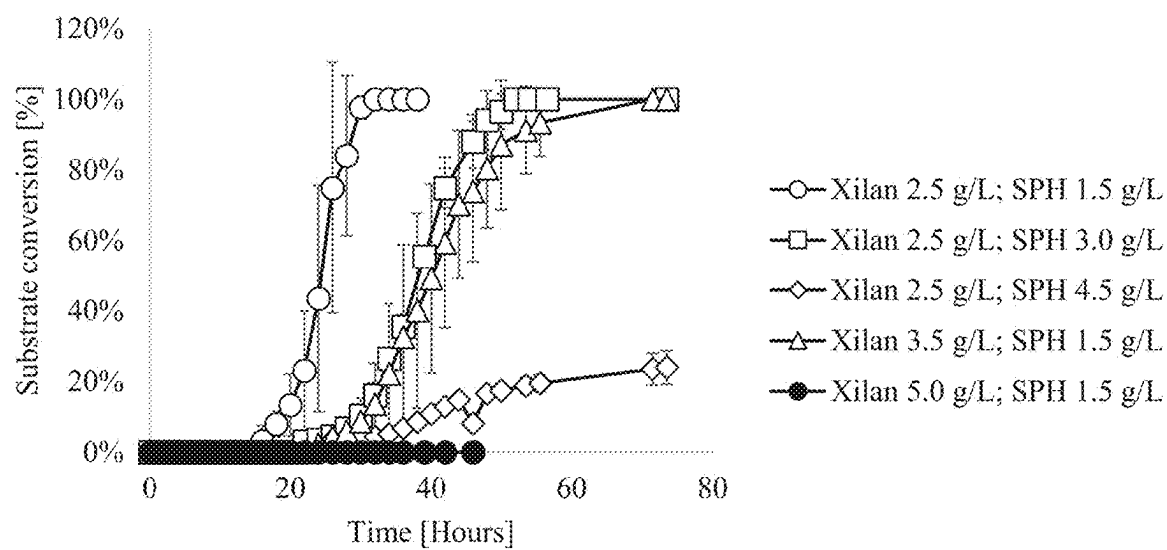


Figure 11

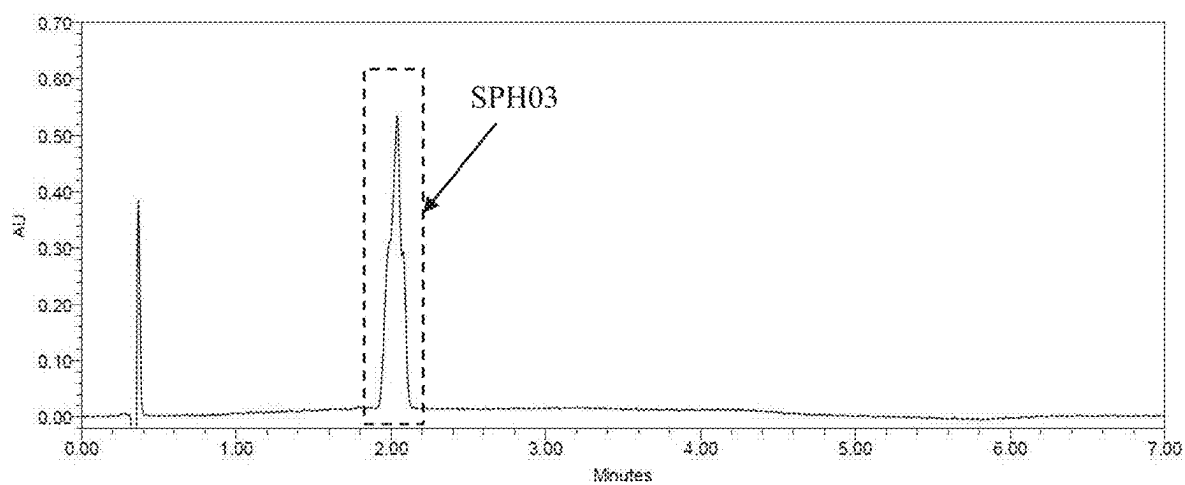


Figure 12

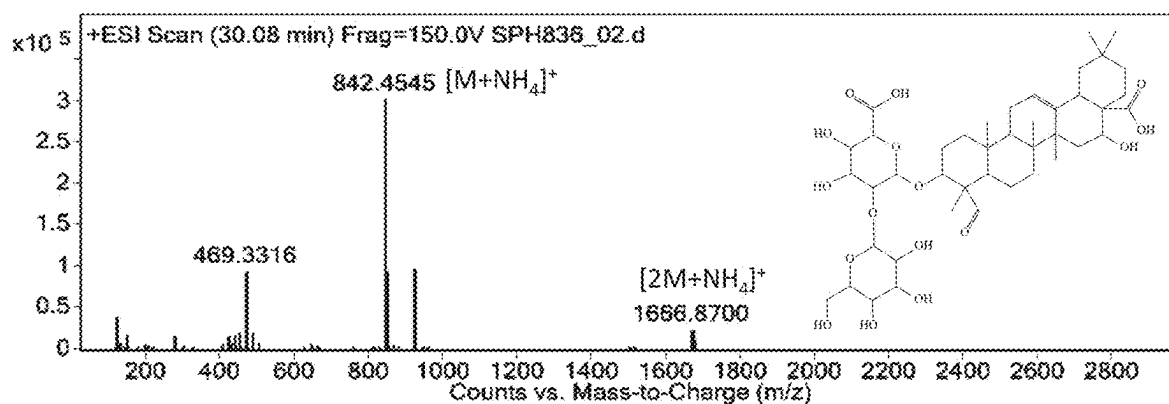


Figure 13

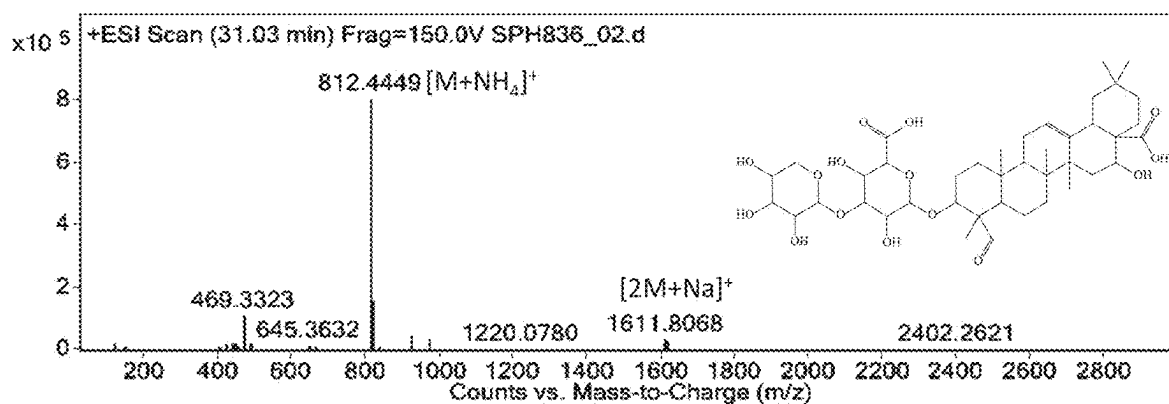


Figure 14

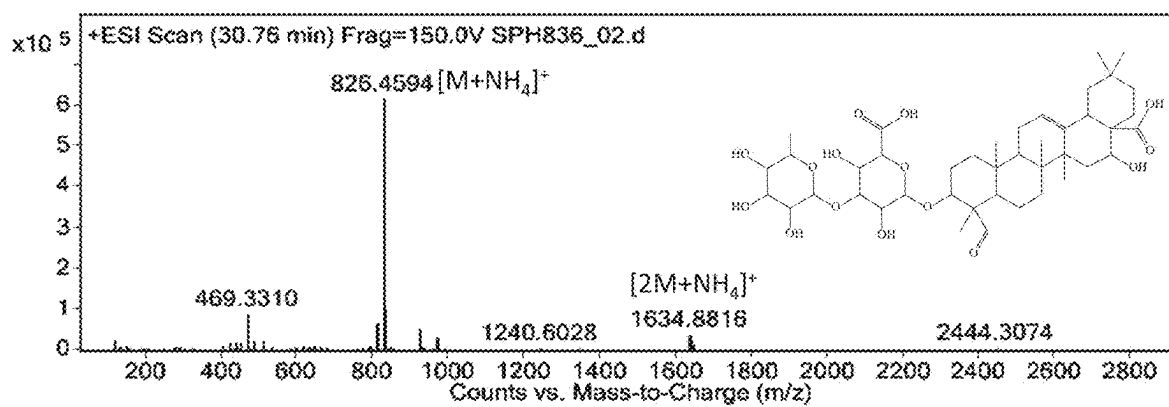


Figure 15

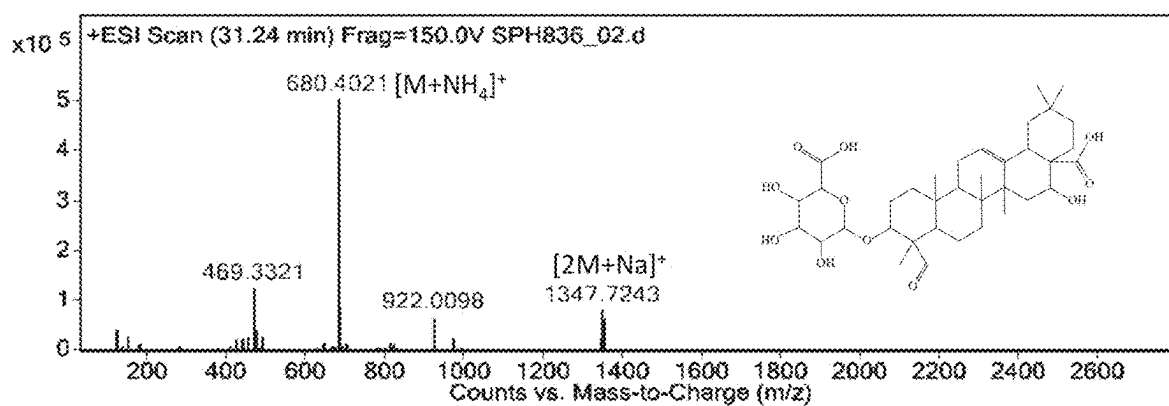


Figure 16A

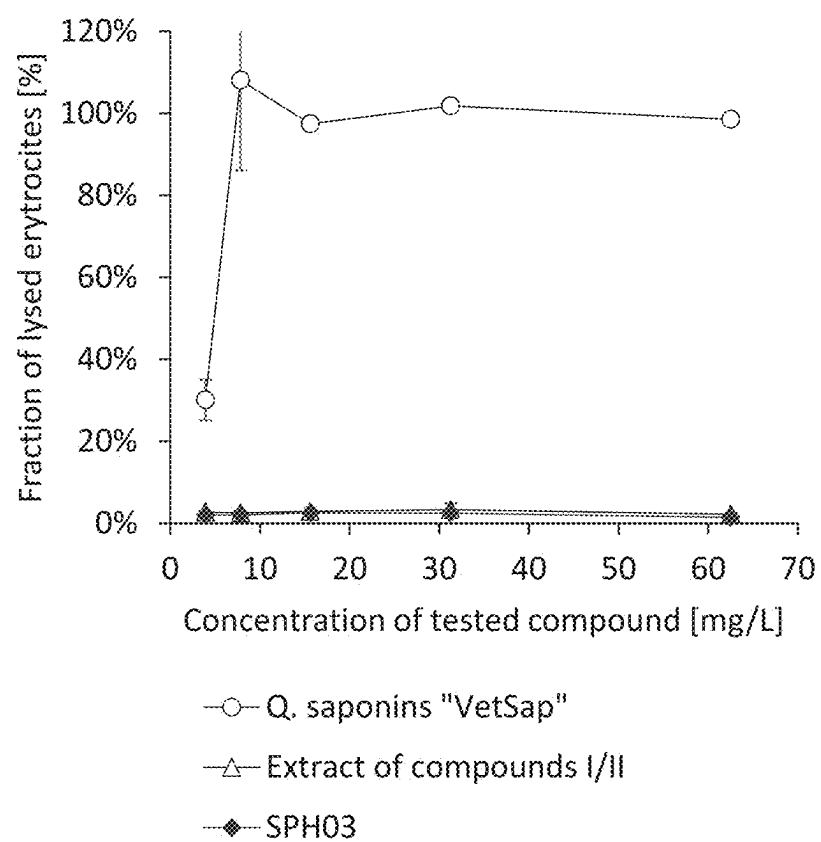


Figure 16B

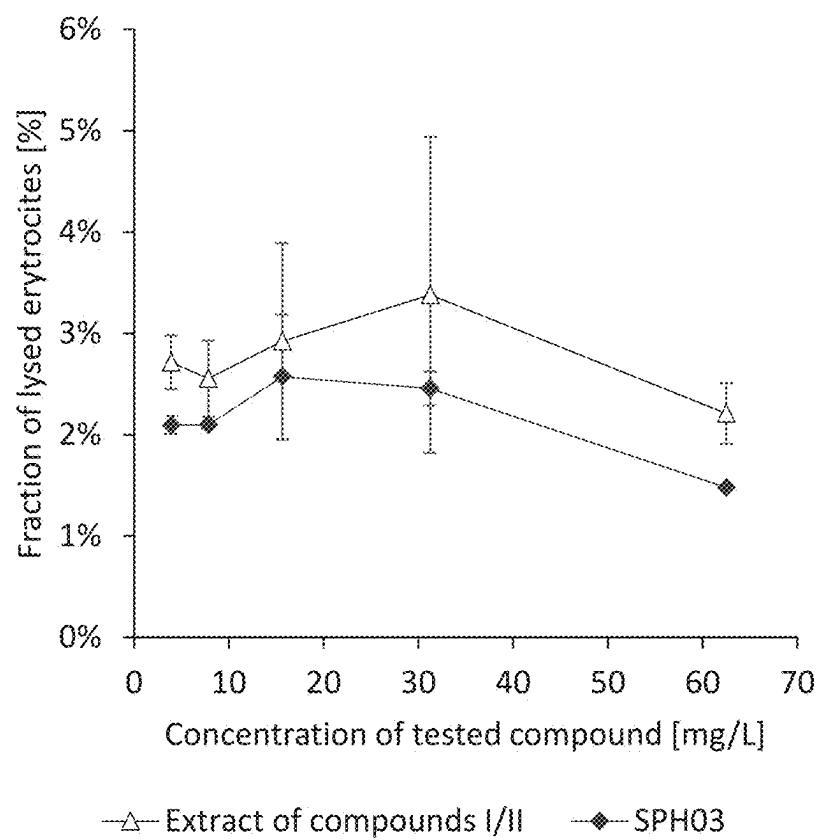


Figure 17

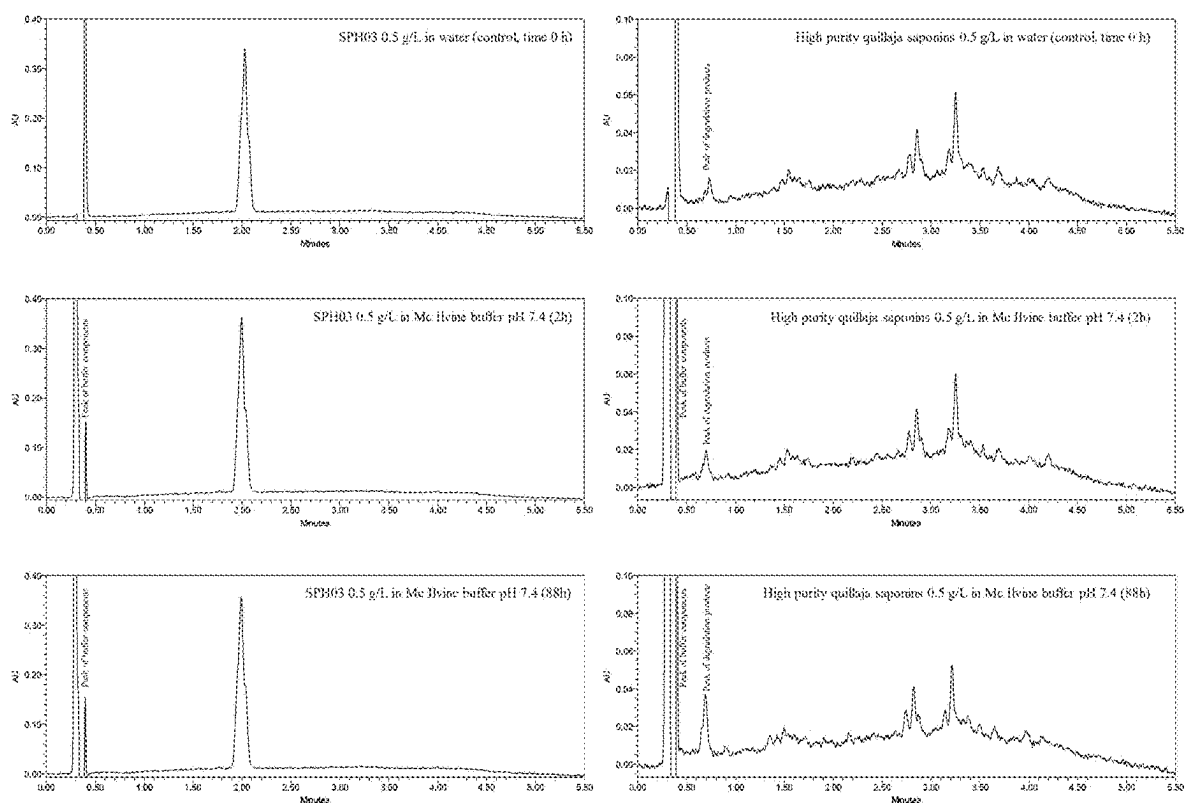


Figure 18A

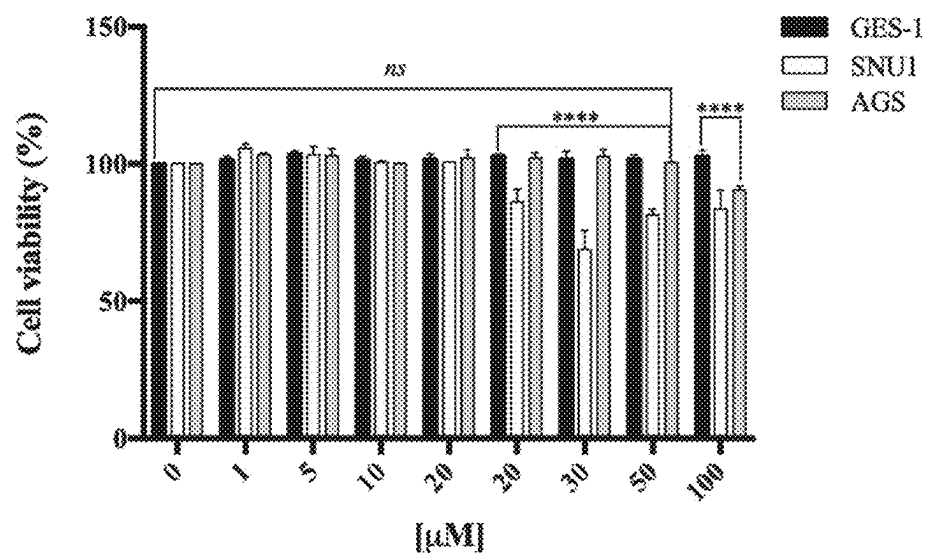


Figure 18B

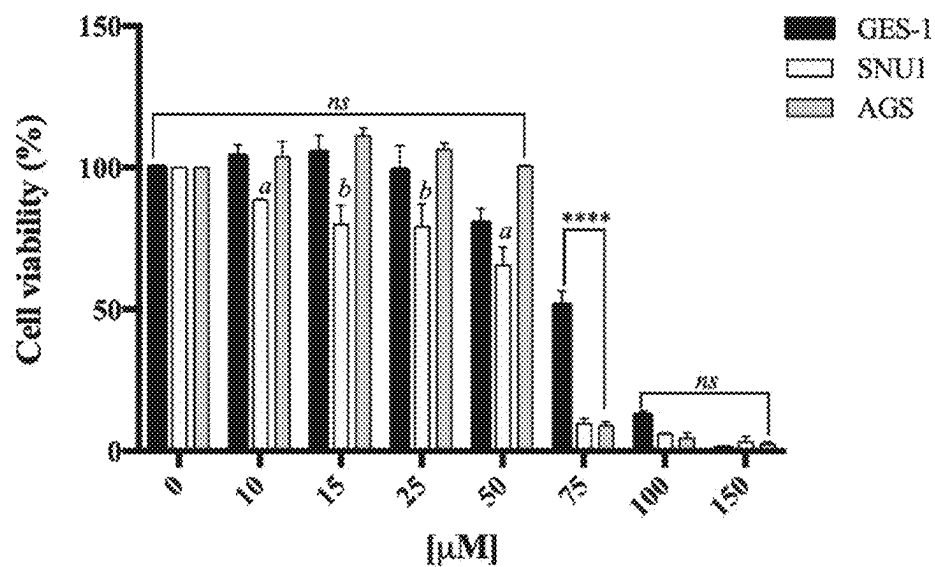


Figure 19

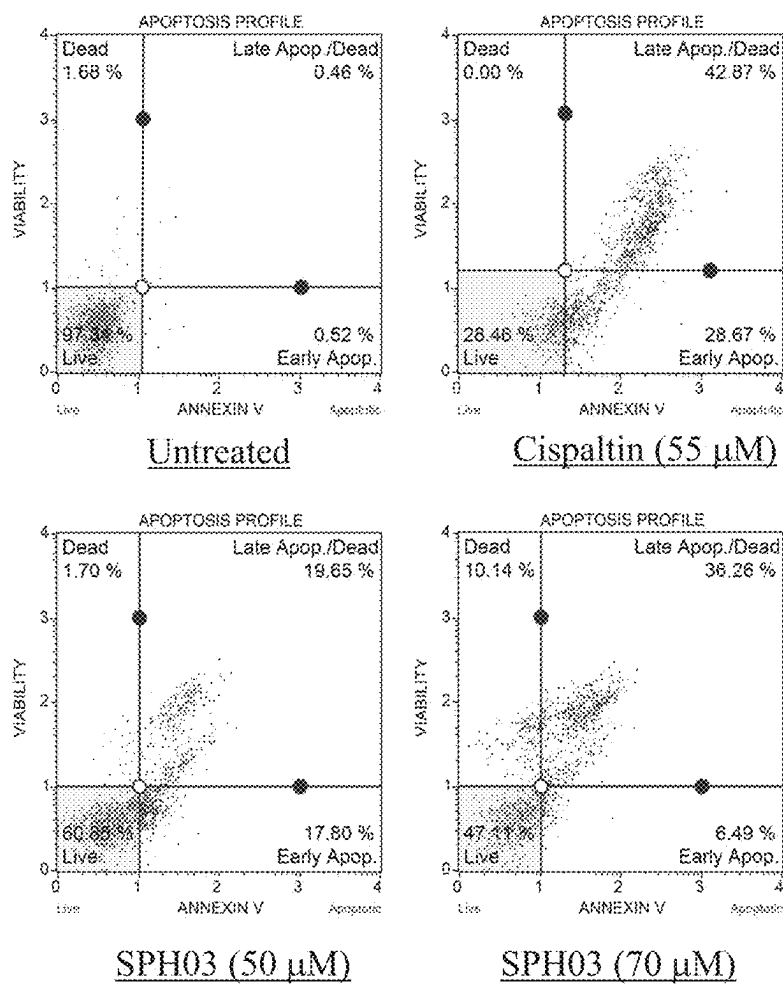


Figure 20

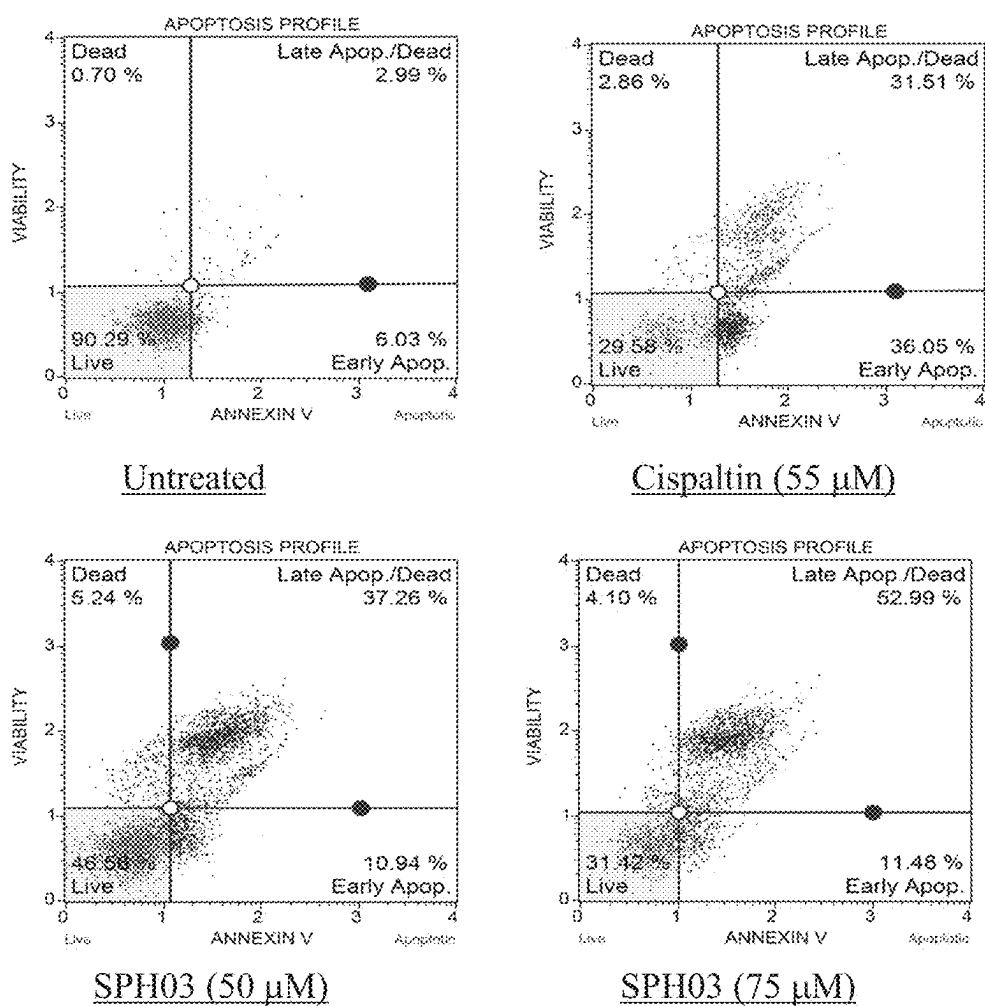
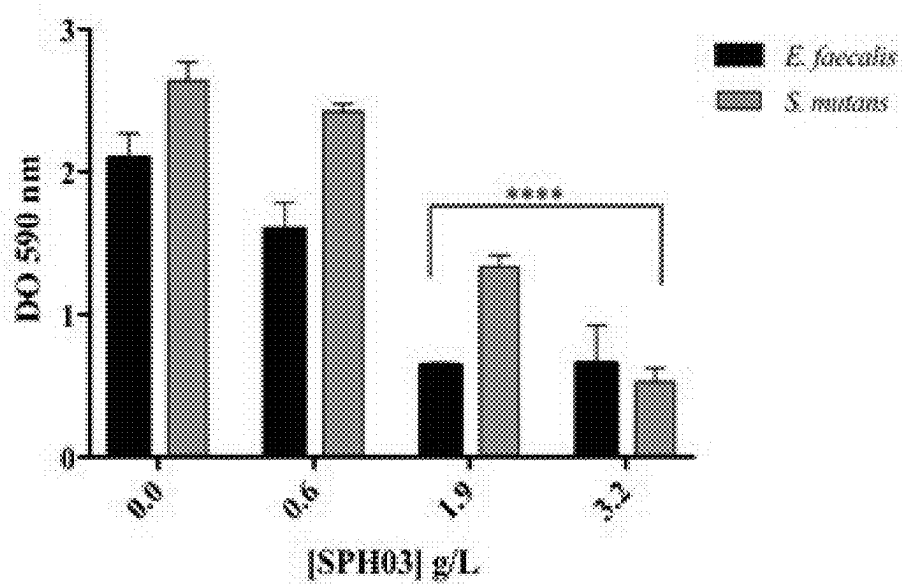


Figure 21



**A 3-O-MONODESMOSIDIC SAPONINS
EXTRACT, STABLE AT PHYSIOLOGICAL
PH, METHOD FOR PREPARING IT FROM A
QUILLAJA PLANT EXTRACT CULTURED
WITH A MICROBIAL CONSORTIUM, USES
AND SUCH MICROBIAL CONSORTIUM**

**AN INCORPORATION BY REFERENCE
STATEMENT REGARDING THE MATERIAL IN**

[0001] An XT.25 file for a “Sequence Listing” (see § 1.831(a), submitted via the USPTO patent electronic filing system, identifying

[0002] the name of the file: Sequence LISTING REVISED;

[0003] the date of creation of each file: Apr. 10, 2025

[0004] and the size of each file in bytes 22 kb bytes.

BACKGROUND OF THE INVENTION

Field

[0005] The present invention relates to a biotransformation process effected by cultures of a microbial consortium for the preparation of monodesmosidic 3-O-glycosides of quillaic acid, also referred as 3-O-monodesmosidic *Quillaja* saponins. Such microbial consortium comprising microorganisms selected from *Rhodotorula* sp having accession number RGM 3031 dated Jul. 22, 2020 (Colony P) of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Microbacterium* sp having accession number RGM 3021 dated Aug. 7, 2020 (Colony Y) of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Mucor* sp having accession number RGM 2984 dated Aug. 7, 2020 (Colony C) of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Paenibacillus* sp and having accession number RGM 3032 dated Sep. 3, 2020 (Colony W) of the Chilean Collection of Microbial Genetic Resources (CChRGM). In particular, the method of this invention provides an extract of 3-O-monodesmosidic saponins prepared from a partially hydrolyzed *Quillaja* plant extract cultured with the above-described microbial consortium.

Description of Related Art

[0006] The biotransformation of saponins has been widely studied and documented for the ginsenosides, the bidesmosidic triterpenic saponins of *Panax ginseng* (*ginseng*). *Panax ginseng* accumulates ginsenosides (Ru et al., *Drug Discovery and Therapeutics*. 9: 23-32 (2015); Shin et al., *Journal of Ginseng Research*. 39: 287-298 (2015)), mostly the compounds Rb1, Rb2, Rc, Rd, Re and Rg1 accounting more than 99% of the saponin composition of *ginseng* (Cheng et al., *Journal of Microbiology and Biotechnology*. 17: 1937-1943 (2007); Kim et al., *Journal of Ginseng Research*. 36: 291-297 (2012)). However, some minor ginsenosides scarcely found in the extracts of *Panax ginseng* have outstanding properties, as for example: i) the ginsenoside M1 (also known as Compound K) has antitumoral, antidiabetic, hepatoprotective and antioxidant properties (Kim et al., 2012; Li et al., *Molecules*. 16: 10093-10103 (2011)); ii) the ginsenoside F2 has antitumoral properties and is also a promising agent to prevent hair-loss (Shin et al., *Journal of Ginseng Research*. 35: 86-92 (2012); Shin et al., *Biological and Pharmaceutical Bulletin*. 37: 755-763 (2014)), and iii)

the compound GP-17 (also known as gipenoside XVII) has potential as neuroprotective agent against the Alzheimer's disease (Meng et al., *Journal of Alzheimers Disease*. 52:1135-1150 (2016)).

[0007] In order to make available large amounts of ginsenoside M1, ginsenoside F2 and the compound GP-17, a number of biotechnological process based in enzymatic conversion and microbial transformation have been described. Li et al., (2011) developed the preparation of the ginsenoside M1 from a mixture of ginsenosides comprising the ginsenosides Rb1, Rb2, Rc y Rd), catalyzed by the enzyme snailase. Cheng et al., (2007) described the preparation of the ginsenoside F and the compound GP-17 by biotransformation of the ginsenoside Rb1 in liquid cultures of the bacterium *Intrasporangium* sp. GS603. Wang et al., (*Molecules*. 20: 19291-19309 (2015)) described the preparation of the ginsenoside M1 by biotransformation of the ginsenoside Rb1 in liquid cultures of the fungus *Cordyceps sinensis*.

[0008] The conversion of saponins by biotechnological methods has been even extended to the preparation of novel compounds from the ginsenosides of *Panax ginseng*. Ko et al., (*Bioscience, Biotechnology and Biochemistry*. 65: 1223-1226 (2001)) described the synthesis of two novel ginsenosides by an enzyme catalyzed reaction between the ginsenoside Re and the glycosyl donors p-nitrophenyl-β-D-xylopyranoside and phenyl-β-D-xylopyranoside. Han et al., (*Molecules*. 15: 399-406 (2010)) prepared the novel antitumoral ginsenoside ORh1 from a mixture comprising ginsenosides Rg1 and Re by a two-step procedure combining enzymatic hydrolysis and chemical acylation.

[0009] Nor of the previously enumerated approaches for bioconversion of saponins has been described for the triterpenic saponins of *Quillaja saponaria* Molina.

[0010] The extracts of the bark and pruned branches of *Quillaja saponaria* Molina contain triterpenic saponins (Resnik, Chemical and Technical Assessment 61st JECFA. FAO (2004)), bidesmosidic glycosides of quillaic acid with two oligosaccharide chains attached to C-3 and C-28 positions (Guo et al., *Phytochemistry*. 48: 175-180 (1998)). The four most abundant saponins in commercial extracts of *Quillaja saponaria* Molina are the bidesmosidic glycosides QS-7, QS-17, QS-18 and QS-21 (Kensil et al., *Journal of Immunology*. 146: 431-437 (1991)).

[0011] The immune-stimulant activity of extracts of triterpenic saponins of *Quillaja saponaria* Molina was first described in the 1930s, and later used to improve a foot-and-mouth disease vaccine. In 1978, Dalsgaard first obtained an enriched mixture of saponins (Quil A) from this extract and found Quil A stimulated both humoral and cellular immunity as well as to induce differential antibody isotypes (Sun et al., *Vaccine*. 27: 1787-1796 (2009)). Since then, the production of several partially purified saponin extracts has been described, for example, SuperSap (Natural Response) and the chromatographically isolated fractions QH-A and QH-C (Rönnberg, et al. *Vaccine*. 13: 1375-1382 (1995)).

[0012] In addition to the immune-stimulant properties, other biological effects have been associated to the triterpenic saponins of *Quillaja saponaria* Molina. The US Patent Application US 2007 0264401 A1, discloses the use of *Quillaja* extracts as beverage preservatives against microbial spoilage by some mold yeast and bacteria. For example, the addition of 100 ppm of *Quillaja* extracts to sucrose

solutions at pH 3.0 stored at 25° C. prevent the growth of the bacteria *Gluconacetobacter xylinus*, *Alicyclobacillus acidoterrestris*, *Bacillus stearothermophilus*.

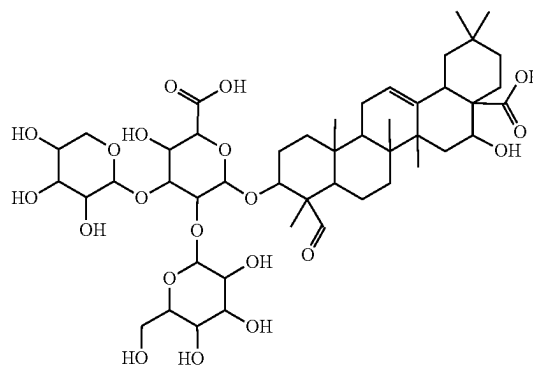
[0013] The US Patent Application US 2005 0175623 A1 discloses the in vitro killing and inhibition effects of a commercial extract of *Quillaja* saponins and some sub-fractions prepared from the said commercial extract on a range of tumor cell lines comprising Hep-G2, HuH-7, HT-29, asPC-1, renal carcinoma cell line, MCF-7, TSU, DMS-53, H-157, Jurkat, and Sp2/0-Ag14. Some of the said sub-fractions of *Quillaja* saponins, alone and in combination with other anti-cancer therapeutic agents showed inhibition of cancer cell growth or proliferation through deconstruction of the cell membrane, and cell cycle arrest at G1, as well as promoting induction of specific cancer apoptosis. The inhibitory effect of saponins observed with cancer cells was not obvious in normal cells such as white blood cells, human hepatocytes, 3T3 mouse cell line and mouse fibroblasts. However, some of the fractions mentioned were shown to be toxic in vivo to mice upon peritoneal and muscular administration. For example, the sub-fraction QSF-IV was also highly toxic to liver (10 mg QSF-IV per Kg of body weight caused 50% death in mice). Although a dose of 3 QSF-IV per Kg of body weight could be injected repeatedly without lethal effects, slight liver damage was still observed. The latter results were consistent with the adverse effects associated with intradermal administration of *Quillaja* saponins to mice, such as hemolysis and lethality (Kensil et al., 1991). To prevent the adverse effects described, as well as to increase the chemical stability of saponins at physiological pH in anticancer applications, the use of liposomes and nanoparticle formulations has been proposed. For example, the US Patent Application US 2010 0119591 A1 discloses the use of lipid containing particles referred as KGI particles comprising cholesterol, phosphatidyl-cholin and *Quillaja* saponin fraction QH-C, as killing agents against malign monoblast cells U937. In addition to their cell killing properties, KGI particles are significantly less hemolytic than free QH-C saponins. According to the related US Patent Application US 2014 0234349 A1, the cell killing effects against U937 cells of nanoparticles G3 comprising only cholesterol and *Quillaja* saponin fraction QH-C, are comparable to those exerted by KGI particles.

[0014] Neither QuilA nor SuperSap are suitable as ingredient of pharmaceutical formulations for human use because of the chemical heterogeneity of the saponin composition, which includes more than 50 different saponins. This chemical heterogeneity has led to the development of procedures for the chromatographic isolation of purified saponins (Kensil & Marciani. U.S. Pat. No. 5,057,540 (1991); Kernsten et al. Canadian Patent 2, 094,600 (1998)). Practical and economic constraints limit the commercial production of purified saponins of *Quillaja saponaria* Molina, and uses thereof, to the most abundant bidesmosidic saponins, for example QS-7, QS-17, QS-18 and QS-21.

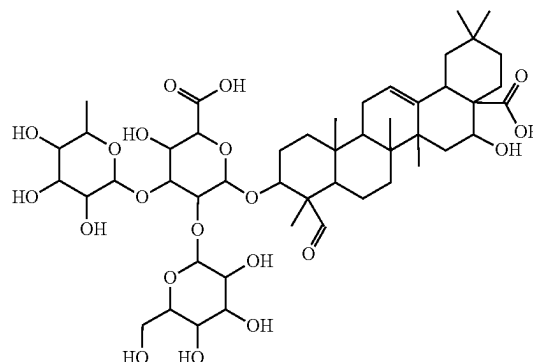
[0015] As previously mentioned, intradermal administration of some bidesmosidic *Quillaja* saponins may cause side effects, because of their hydrophobic-lytic properties resulting in trapping at the site of administration, causing cell and tissue destruction, leading to local and systemic adverse reactions (Hu et al., *International Journal of Nanomedicine*. 5: 51-62 (2010)). These adverse effects have prevented the widespread use of saponins for injectable administration in therapeutic applications.

[0016] An additional major concern related to the use of bidesmosidic saponins of *Quillaja saponaria* Molina is the lack of chemical stability at physiological pH, leading to the hydrolysis of the ester bond joining the substituent at C-28 position (Pillion et al. *Journal of Pharmaceutical Sciences*. 85: 518-524 (1996)).

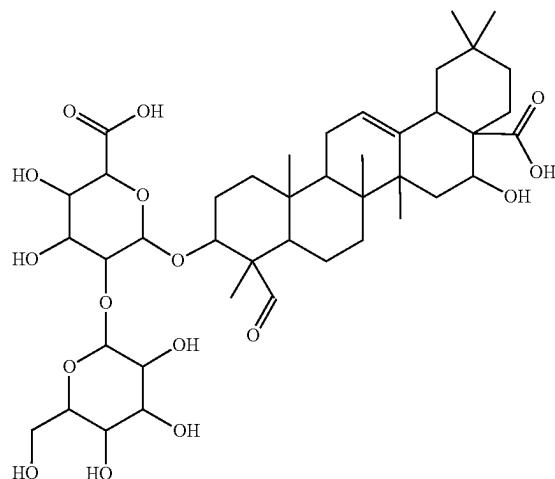
[0017] Thus, there is an unmet need of stable *Quillaja* saponins lacking the above adverse effects. The monodesmosidic *Quillaja* saponins lacking the substituents at C-28 are scarcely found in *Quillaja saponaria* Molina (Guo et al., *Phytochemistry*. 48: 175-180 (1998)). Guo et al., 1998 purified from the QH-A fraction of extracts of *Quillaja saponaria* three monodesmosidic 3-O-glycosides of quillaic acid, chemically characterized as quillaic acid 3-O-{xylopyranosyl-(1→3)-[galactopyranosyl-(1→2)]-glucopyranosiduronic acid}



hereinafter referred as Compound I, quillaic acid 3-O-{rhamnopyranosyl-(1→3)-[galactopyranosyl-(1→2)]-glucopyranosiduronic acid}



hereinafter referred as Compound II and quillaic acid 3-O-{rhamnopyranosyl-(1→2)-glucopyranosiduronic acid}



hereinafter referred as Compound III.

[0018] According to the data of Guo et al., 1998 the yields of the recovery of the Compounds I, II and III from QH-A fraction by chromatographic means were 2.24%, 0.71%, and 0.29%, respectively.

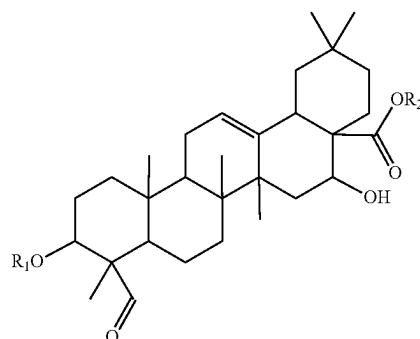
[0019] The U.S. Pat. No. 5,650,398 discloses a method to produce the compound QH-957 having the same structure of the Compound I, by heating a QS-21 solution in phosphate-buffered saline at pH 6-7 for 48-72 hours at 100°-105° C., followed by purification of Compound I by reverse-phase HPLC in a water/acetonitrile gradient, containing 0.15% trifluoroacetic acid, and lyophilization of the corresponding fractions.

[0020] According to Van Setten et al., (*Rapid Communications in Mass Spectrometry*, 9: 660-666 (1995)) the base catalyzed hydrolysis of the partially purified extract of QuilA with KOH in 50% ethanol at reflux temperature for 7 hours, yielded only two main products with molecular masses 956 and 970 D, consistent with the molecular masses of the anionic forms of Compounds I and II, respectively; the production of the Compound III under these conditions was not detected.

[0021] Moreover, the U.S. Pat. No. 5,817,314 discloses the purification by chromatographic means from extracts of the bark of *Quillaja saponaria* Molina, of a compound QS-L1 having the same molecular mass of the anionic form of Compound I, which possess immune-stimulant properties and no hemolytic activity in the range 4-500 µg/ml. However, the isolation of the compound QS-L1 described in the latter disclosure still depends on the supply of bark extracts to secure the isolation of the compound QS-L1. Thus, there is an unmet need to find novel methods to produce 3-O-monodesmosidic saponins of *Quillaja saponaria* Molina suitable for pharmaceutical formulations in greater yield.

SUMMARY OF THE INVENTION

[0022] An object of the present invention is to provide an extract of saponin compounds, hereafter designated as extract SPH03, comprising one or more 3-O-monodesmosidic saponins of Formula (I),



[I]

wherein R₁ is {xylopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, including {xylopyranosyl-(1→3)-glucopyranosiduronate salt}; {rhamnopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, including {rhamnopyranosyl-(1→3)-glucopyranosiduronate salt}; or {galactopyranosyl-(1→2)-glucopyranosiduronic acid} or a salt thereof, including {galactopyranosyl-(1→2)-glucopyranosiduronate salt}; or {glucopyranosiduronic acid} or a salt thereof, {glucopyranosiduronate salt}, and R₂ is hydrogen or a cation specie.

[0023] The 3-O-monodesmosidic saponins of Formula [I] (as described above) in the extract SPH03 are stable at physiological pH, have no or little hemolytic effect, i.e., a strong reduce hemolytic effect, and exhibit, cytotoxic effect on cancer cells and antibacterial effect on oral pathogens.

[0024] Another object of the present invention is to provide a novel microbial consortium [HNR2] for the microbial biotransformation of *Quillaja* plant extract comprising Compounds I and II, to the extract SPH03. Such microbial consortium comprising microorganisms selected from *Rhodotorula* sp having accession number RGM 3031 (Colony P) dated Jul. 22, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Microbacterium* sp having accession number RGM 3021 (Colony Y) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Mucor* sp having accession number RGM 2984 (Colony C) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Paenibacillus* sp and having accession number RGM 3032 (Colony W) dated Sep. 3, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM).

[0025] Still another object of this invention is to provide a biotransformation process to overcome the limitations for production of large amounts of the extract SPH03 substantially free of bidesmosidic saponins of *Quillaja* plants, preferably *Quillaja saponaria* Molina extract, based on liquid fermentation cultures of HNR2 microbial consortium catalyzing the conversion of an extract comprising Compounds I and II, to the extract SPH03, wherein such microbial consortium is as mentioned above.

[0026] A further object of this invention is to provide a formulation of the liquid broth employed for the biotransformation of an extract comprising Compounds I and II, to the extract SPH03 catalyzed by HNR2, where the productivity of the process is increased.

[0027] A still further object of this invention is to provide a method to isolate the extract SPH03 from the fermentation broth employed in the biotransformation.

[0028] These objects are accomplished by the herein described invention which comprises steps of hydrolysis of aqueous extracts of *Quillaja* plant extract, preferably, a *Quillaja saponaria* Molina extract, followed by the biotransformation of the resulting hydrolyzed extract comprising Compounds I and II by a HNR2 microbial consortium culture as mentioned above, and the subsequent recovery of the extract SPH03 substantially free of bidesmosidic saponins of *Quillaja saponaria* Molina from the fermentation broth by liquid extraction or chromatographic means. The method of the present invention allows the practical, reliable and sustainable production of large amounts of the extract SPH03 as well as the 3-O-monodesmosidic saponins of Formula (I).

[0029] This invention also provides examples of the use of the extract SPH03 as well as its monodesmosidic saponins of Formula (I) as cytotoxic and antibacterial agents.

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1: RP-UHPLC trace of VaxSap® (VS, highly purified powder extract, mainly containing triterpene saponins ca. 90% w/w), a spray-dried and highly purified extract of saponins of *Quillaja saponaria* Molina.

[0031] FIG. 2: The conversion of *Quillaja saponaria* saponins (VaxSap®) to a mixture comprising Compounds I and II, as verified by RP-UHPLC chromatography.

[0032] FIG. 3: TLC analysis of the eluate fractions after the hydrolysis of *Quillaja saponaria* saponins (VaxSap®).

[0033] FIG. 4: Gel like substance floating at the surface of a spoiled aqueous extract of *Quillaja saponaria* Molina stored at room temperature, employed as the source for the isolation of the consortium HNR2 (as defined above).

[0034] FIG. 5: Four major types of colonies (*Mucor* sp (RGM 2984, C), *Rhodotorula* sp (RGM 3031, P), *Paenibacillus* sp (RGM 3032, W) and *Microbacterium* sp (RGM 3021, Y) found upon plating of inoculum HNR2 on Luria Bertani Agar (LBA).

[0035] FIG. 6A: Agarose electrophoresis of PCR products obtained from genomic DNA of colonies C, P, W and Y, with primer pairs PP1 (SEQ ID Nos.: 1 and 2).

[0036] FIG. 6B: Agarose electrophoresis of PCR products obtained from genomic DNA of colonies C, P, W and Y, with primer pairs PP2 (SEQ ID Nos.: 3 and 4).

[0037] FIG. 6C: Agarose electrophoresis of PCR products obtained from genomic DNA of colonies C, P, W and Y, with primer pairs PP3 (SEQ ID Nos.: 5 and 6).

[0038] FIG. 6D: Agarose electrophoresis of PCR products obtained from genomic DNA of colonies C, P, W and Y, with primer pairs PP4 (SEQ ID Nos.: 7 and 8).

[0039] FIG. 7: RP UHPLC analysis of supernatants taken during the biotransformation of Compounds I and II to SPH03 catalyzed by consortium HNR2 as defined above.

[0040] FIG. 8: TLC analysis of supernatants taken during the biotransformation of Compounds I and II to SPH03 catalyzed by consortium HNR2 as defined above.

[0041] FIG. 9: TLC analysis of supernatants taken during the biotransformation of Compounds I and II to SPH03 catalyzed by consortium HNR2 as defined above, showing the effect of the broth formulation on the production of SPH03.

[0042] FIG. 10: Effect of xylan concentrations and Compounds I and II on the conversion of an extract containing Compounds I and II in SPH03 by liquid cultures of consortium HNR2 as defined above.

[0043] FIG. 11: RP UHPLC chromatogram of the SPH03 product purified by preparative RP HPLC.

[0044] FIG. 12: Mass spectrum of the Compound III of the extract SPH03, obtained by high performance liquid chromatography coupled to electrospray ionization time of flight mass spectrometry analysis (HPLC-ESI-TOF-MS).

[0045] FIG. 13: Mass spectrum of the Compound IV of the extract SPH03, obtained by HPLC-ESI-TOF-MS analysis.

[0046] FIG. 14: Mass spectrum of the Compound V of the extract SPH03, obtained by HPLC-ESI-TOF-MS analysis.

[0047] FIG. 15: Mass spectrum of the Compound VI of the extract SPH03, obtained by HPLC-ESI-TOF-MS analysis.

[0048] FIG. 16A: Hemolytic effect on chicken red blood cells (CRBC) of *Quillaja* saponins (VetSap, Veterinary Vaccine Adjuvant from a semi-purified Saponin Fraction of the Bark of the *Quillaja Saponaria* tree), an extract of compounds I/III and SPH03.

[0049] FIG. 16B: Hemolytic effect on chicken red blood cells (CRBC) of an extract of compounds I/III and SPH03.

[0050] FIG. 17: UHPLC analysis of samples of 0.5 g/L solutions of *Quillaja* saponins and SPH03 in Mc Ilvine buffer (pH 7.4) incubated at 37° C.

[0051] FIG. 18A: Effect of the extract comprising Compounds I and II on cell viability tumoral cells (AGS, SNU1 and GES-1 cells cultured in vitro, as determined by MTS assay, a colorimetric cell proliferation assay).

[0052] FIG. 18B: Effect of SPH03 on cell viability tumoral cells (AGS, SNU1) and GES-1 cells cultured in vitro, as determined by MTS assay.

[0053] FIG. 19: Muse™ Annexin V and Dead Cell apoptosis assay applied to AGS cells untreated, treated with Cisplatin 55 microM, SPH03 50 microM and SPH03 70 microM.

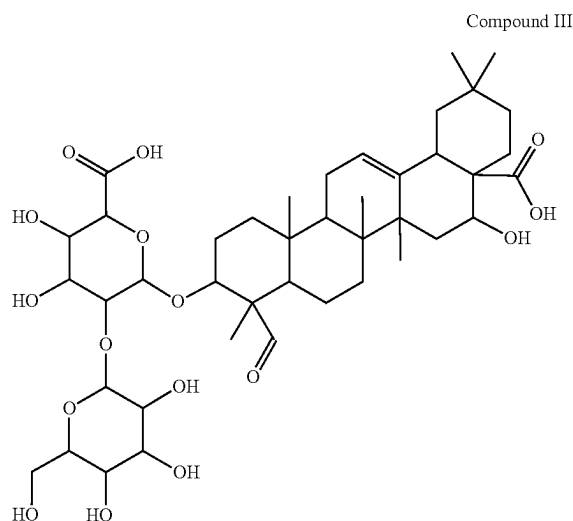
[0054] FIG. 20: Muse™ Annexin V and Dead Cell apoptosis assay applied to SNU1 cells untreated, treated with Cisplatin 55 microM, SPH03 50 microM and SPH03 70 microM.

[0055] FIG. 21: Inhibition of biofilm formation by *Streptococcus mutans* UA159 and *Enterococcus faecalis* ATCC 29212 grown in BHIB broth supplemented with SPH03.

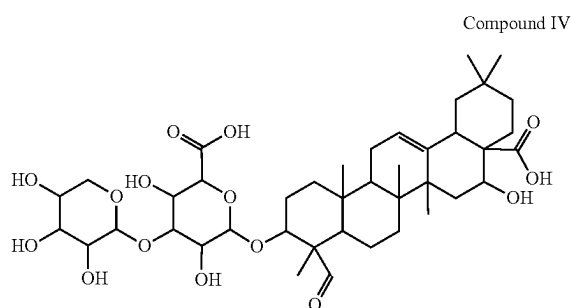
DETAILED DESCRIPTION

[0056] As described above, the present application relates to a method for obtaining an *Quillaja* plant extract comprising monodesmosidic 3-O-glycosides of quillaic acid, being stable at physiological pH, have no or little hemolytic effect, i. e., a strong reduced hemolytic effect, and exhibit, cytotoxic effect on cancer cells and antibacterial effect on oral pathogens such as *Streptococcus mutans*.

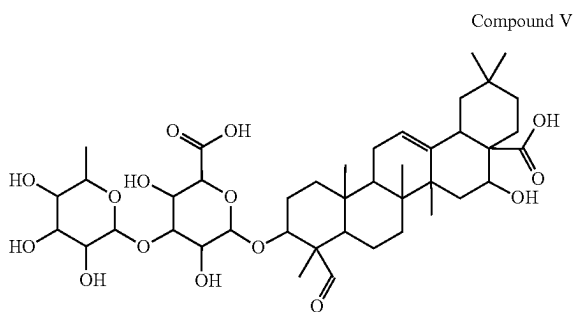
[0057] Referring now to the invention in more detail, the extract SPH03, comprising one or more of the monodesmosidic saponins of Formula (I), herein below referred as Compounds III (as described above), IV, V and VI, are prepared by biotransformation of a plant biomass extract comprising glycosides of quillaic acid with liquid fermentation cultures of HNR2 microbial consortium, being compounds III, IV, V and VI as follows:



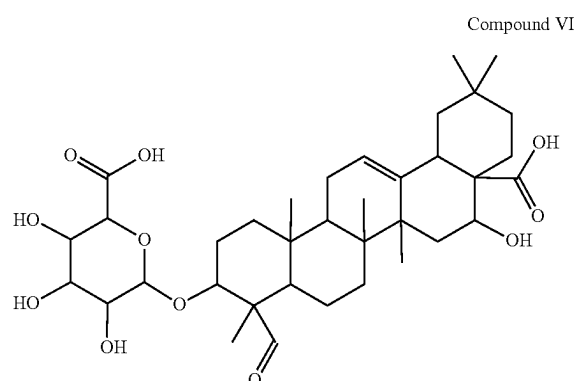
[0058] Compound III: Quillaic acid 3-O-{galactopyranosyl-(1→2)-glucopyranosiduronic acid} or a salt thereof, quillaic acid 3-O-{galactopyranosyl-(1→2)-glucopyranosiduronate salt} and R_2 is hydrogen or a cation specie.



[0059] Compound IV: Quillaic acid 3-O-{xylopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, quillaic acid 3-O-{xylopyranosyl-(1→3)-glucopyranosiduronate salt} and R_2 is hydrogen or a cation specie.



[0060] Compound V: Quillaic acid 3-O-{rhamnopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, quillaic acid 3-O-{rhamnopyranosyl-(1→3)-glucopyranosiduronate salt} and R_2 is hydrogen or a cation specie.



[0061] Compound VI: Quillaic acid 3-O-glucopyranosiduronic acid or a salt thereof, quillaic acid 3-O-glucopyranosiduronate salt and R_2 is hydrogen or a cation specie.

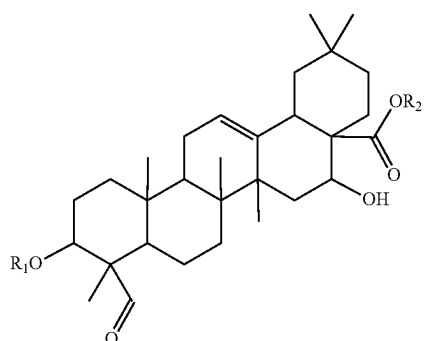
[0062] Such microbial consortium was taken out from spoiled quillaja extracts. Table 1 below summarizes the characteristics of the microorganisms forming the consortium as well as its deposit-related information

TABLE 1

Name of the No microorganism.	Characteristics of the microorganism.	Deposit No in CChRGM
1 <i>Paenibacillus</i> sp. (W)	Facultative anaerobic, endospore-forming bacteria.	RGM 3032 Sep. 3, 2020
2 <i>Microbacterium</i> sp. (Y)	<i>Microbacterium</i> is a genus of bacteria in the family Microbacteriaceae.	RGM 3021 Aug. 7, 2020
3 <i>Rhodotorula</i> sp. (P)	Common environmental basidiomycetous yeasts, which can be found in soil, ocean and lake water, fruit juice and milk.	RGM 3031 Jul. 22, 2020
4 <i>Mucor</i> sp. (C)	<i>Mucor</i> is a microbial genus of approximately 40 species of molds commonly found in soil, digestive systems, plant surfaces, foods and rotten vegetable matter.	RGM 2984 Aug. 7, 2020

[0063] The extract SPH03, comprising one or more of the monodesmosidic saponins of Formula (I) as defined above, which are referred below as Compounds III, IV, V and VI, being stable at physiological pH, having no or little hemolytic effect, i.e, a strong reduced hemolytic effect, and exhibit anti-tumor effect on cancer cells and antibacterial effect on oral pathogens, such as *Streptococcus mutans*.

[0064] Thus, the term “extract SPH03” is used herein below to refer an extract of saponin compounds comprising one or more 3-O-monodesmosidic saponins of Formula (I),



wherein R_1 is {xylopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, including {xylopyranosyl-(1→3)-glucopyranosiduronate salt}; {rhamnopyranosyl-(1→3)-glucopyranosiduronic acid} or a salt thereof, including {rhamnopyranosyl-(1→3)-glucopyranosiduronate salt}; or {galactopyranosyl-(1→2)-glucopyranosiduronic acid} or a salt thereof, including {galactopyranosyl-(1→2)-glucopyranosiduronate salt}; or {glucopyranosiduronic acid} or a salt thereof, {glucopyranosiduronate salt}, and R_2 is hydrogen or a cation specie, which is substantially free of bidesmosidic saponins.

[0065] The term “HNR2 microbial consortium” is used herein to refer a microbial consortium comprising microorganisms selected from *Rhodotorula* sp having accession number RGM 3031 (colony P) dated Jul. 22, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Microbacterium* sp having accession number RGM 3021 (colony Y) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Mucor* sp (colony C) having accession number RGM 2984 dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Paenibacillus* sp and having accession number RGM 3032 (colony W) dated Sep. 3, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM).

[0066] The term “hemolytic effect” as used herein refers to the lysis of erythrocyte cells.

[0067] The term “cytotoxic effect” as used herein refers to the inhibition of the replication of cancer cells and the decrease in the number of cancerous cells in vitro or in vivo.

[0068] The term “antibacterial effect” as used herein refers to the inhibition of the replication of bacterial cells and the decrease in the number of bacterial cells in vitro or in vivo.

[0069] The term “plant biomass” as used herein refers to as used herein, refers to any biological material originated from the kingdom Plantae. For example, the biomass can be the bark, trunk, leaves, stems, roots, seeds, flowers, fruits or a combination thereof.

[0070] The term “plants containing glycosides of quillaic acid” as used herein refers, but not limited to, the following plant species: *Quillaja saponaria* Molina, *Quillaja brasiliensis* and *Saponaria vaccaria*.

[0071] The present invention provides a method for production of the extract SPH03 substantially free of bidesmosidic saponins as defined above, comprising the following steps:

[0072] a) Preparing a partially hydrolyzed extract from a plant biomass extract containing glycosides of quillaic acid and obtaining a crude hydrolysate,

[0073] b) Recovering from the crude hydrolysate obtained in step a), an extract of monodesmosidic saponins comprising Compounds I and II,

[0074] c) Culturing such extract obtained in step b) into a liquid fermentation culture of a microbial consortium (HNR2), wherein such microbial consortium (HNR2) comprising microorganisms selected from *Rhodotorula* sp having accession number RGM 3031 (colony P) dated Jul. 22, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Microbacterium* sp having accession number RGM 3021 (colony Y) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Mucor* sp having accession number RGM 2984 (colony C) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Paenibacillus* sp and having accession number RGM 3032 (colony W) dated Sep. 3, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM), and

[0075] d) Removing such microbial consortium (HNR2) from such fermentation broth obtained in step c) and recovering an extract substantially free of microorganisms and bidesmosidic saponins.

[0076] In preferred embodiments, the plant biomass for the preparation of the extract SPH03 is selected from bark, pruned branches or leaves of a *Quillaja* plant, preferably, from *Quillaja saponaria* Molina.

[0077] Such extract of biomass can be preferably hydrolyzed in basic media, with an inorganic or organic base, prior to transform the bidesmosidic saponins in 3-O-monodesmosidic saponins (Compounds I and II).

[0078] In a preferred embodiment, the extract of biomass of *Quillaja saponaria* Molina can be preferably hydrolyzed in KOH solutions and then also submitted to aqueous ethanol to render a crude hydrolysate comprising Compounds I and II as well as other impurities.

[0079] Such crude hydrolysate can be also fractionated rendering an enriched extract comprising Compounds I and II. In a preferred embodiment the crude hydrolysate is fractionated by chromatography on a hydrophobic resin.

[0080] Such enriched extract comprising Compounds I and II can be also added to the culture broth inoculated with a consortium of biotransforming microorganisms. Such microbial consortium can also comprise one or more of *Paenibacillus amylolyticus*, *Paenibacillus tundra*, *Paenibacillus tylopili*, *Microbacterium hydrocarbonoxydans*, *Microbacterium luteolum*, *Microbacterium phyllosphaerae*, *Rhodotorula mucilaginosus*, *Mucor reemosus*, *Cylindrocarpon* spp., *Aspergillus niger* and *Aspergillus parasiticus*. Preferably, such consortium is consortium HNR2 as defined above.

[0081] The culture broth supplemented with the enriched extract comprising Compounds I and II and inoculated with the consortium of biotransforming microorganisms (HNR2) is cultivated rendering the biotransformed fraction SPH03 comprising the Compounds III, IV, V and VI. To improve the production of SPH03 by the consortium of biotransforming microorganisms, the culture broth can be further supplemented with oligosaccharides of xylose.

[0082] The extract SPH03 containing Compounds III, IV, V and VI may be isolated from the biotransformation broth

by solvent extraction followed by solvent evaporation at reduced pressure, or by chromatography followed by solvent removal at reduced pressure and lyophilization.

[0083] Compounds HI, IV, V and VI of the extract SPH03, may be identified by a pseudo-molecular ion of the mono-sodium adducts when analyzed by mass spectrometry using electrospray ionization (ESI-MS).

[0084] The advantages of the present invention include, without limitation, the production of the extract SPH03, substantially free of bidesmosidic saponins. Other advantage is the production of the extract SPH03 of a range of plant biomass. The present method may be further understood by reference to the following Examples, which are not intended to be limiting of the scope of the invention.

EXAMPLES

Example 1: Analysis of Saponin Fractions by UHPLC Liquid Chromatography

[0085] The chemical profile of the saponins present in substrate and product solutions of biotransformation steps was determined by ultrahigh performance liquid chromatography (UHPLC) in a Waters Acquity H-Class UPLC chromatography system coupled to a Waters PDA Acquity diode array absorbance detector. The analytical column used was an Acquity UPLC® BEH C-18 (particle size 1.7 µm, inner diameter 2.1 mm; length 50 mm) (Waters, MA, USA). The column was maintained at 30° C. during each chromatographic run. The injected samples were eluted at constant flow rate (0.41 ml/min) in gradient elution mode employing the following mobile phases: Solvent A: Formic acid 0.15% v/v in water, and; Solvent B: Formic acid 0.15% v/v in acetonitrile. The solvent gradient employed is described as follows: Step 1: 0.0-2.5 min from 34% to 45% of Solvent B; Step 2: 2.5-3.5 min 45% of Solvent B; Step 3: 3.5-5.0 min from 45% to 34% of Solvent B, and; Step 4: 5.0-7.0 min 34% of Solvent B. The eluted peaks were detected by monitoring of the absorbance at 210 nm.

Example 2: Characterization of the Products of Biotransformation by Liquid Chromatography Coupled to Mass Spectrometry

[0086] The triterpenes present in the biotransformation product were characterized by high performance liquid chromatographic methods using “time of flight” mass detection (HPLC-ESI-TOF-MS) in an Agilent HPLC 1200 HPLC chromatography system coupled to an Agilent Q-TOF 6520 mass spectrometer detector. The column used was a Luna® 5 µC18 (particle size 5.0 µm, inner diameter 4.6 mm; length 250 mm) (Phenomenex, CA, USA). The column was maintained at 30° C. during each run. The injected samples were eluted in gradient elution mode at constant flow rate (1.0 ml/min) employing the following mobile phases: Solvent A: Formic acid 0.10% v/v in water, and; Solvent B: Formic acid 0.10% v/v in acetonitrile. The solvent gradient employed is described as follows: Step 1: 0-30 min from 34% to 45% of Solvent B; Step 2: 30-35 min 45% of Solvent B; Step 3: 35-36 min from 45% to 34% of Solvent B, and; Step 4: 36-50 min 34% of Solvent B.

[0087] The eluted compounds were ionized in electrospray positive ionization mode. The mass detector parameters were as follows: Capillary potential: 4.0 kV; fragmentor voltage: 150 V, and; mass range: 100-3000 Th.

Example 3: Analysis of Saponin Fractions by Thin-Layer Chromatography (TLC)

[0088] In addition to liquid chromatography, thin-layer chromatography (TLC) was employed to characterize the composition of substrates and biotransformation products. To this end 2-4 microliters of sample (25 g/L) were loaded onto aluminum TLC plates covered with Silicagel 60. The plate was developed with a mixture 30:16:8:1 of ethyl acetate/ethanol/water/acetic acid. After this procedure, the plate was air-dried and sprayed with a visualization reagent prepared by mixing of 48 ml ethanol, 2 ml deionized water, 2.24 ml p-anisaldehyde and 1.35 ml concentrated sulfuric acid. The sprayed plate was heated 10 min at 105° C. to visualize the bands.

Example 4: Preparation of an Extract Containing Compounds I and II, Employed as Substrate of the Biotransformation Process

[0089] A spray-dried and highly purified extract of saponins of *Quillaja* (VaxSap®) was employed as starting material for the preparation of an extract containing Compounds I and II from *Quillaja saponaria* Molina, FIG. 1 shows a representative RP-UHPLC chromatogram of Vax-Sap.

[0090] Twenty (20) grams of VaxSap® were suspended in 1000 ml of KOH 28 g/L in aqueous ethanol (50% v/v) and heated at reflux temperature for 5 hours. Once completed the reflux, the pH of the resultant solution was adjusted to the range 6-7 with hydrochloric acid 6M rendering the crude saponin hydrolysate. After drying of 1 L of crude hydrolysate at 50° C. in a heating oven, 48.2 g dried crude saponin hydrolysate were recovered. The conversion of *Quillaja saponaria* saponins to a mixture containing Compounds I and II was verified by RP-UHPLC chromatography, as shown in FIG. 2.

[0091] The dried crude saponin hydrolysate was fractionated by column chromatography on a hydrophobic vinyl-divinyl benzene resin. In a representative example, 20 g of hydrophobic vinyl-divinyl benzene resin (Diaion HP-20 resin), were soaked in approximately 2.5 volumes of absolute ethanol for 24 hours. The resultant slurry was transferred to a glass column supplied with a stopcock valve (diameter 1.9 cm; length 30 cm). The column bed (approximately 51 ml) was washed with 250 ml of deionized water. The dried crude saponin hydrolysate (6.4 g) was dissolved in 64 ml of deionized water and acidified to pH 3 by addition of 1.28 ml of formic acid. The resultant solution was fed to the column at an approximate flow rate of 5 ml/min. The loaded crude saponin hydrolysate was fractionated by serial elution with 150 ml of water, 150 ml of aqueous ethanol 40% v/v, 150 ml aqueous ethanol 80% v/v and 150 ml of absolute ethanol. The relative content of Compounds I and II in each eluate was verified by TLC chromatography following the procedure described in Example 3. According to the results of the TLC test, the eluate fraction having the highest content of Compounds I and II was the eluate collected after elution with aqueous ethanol 80% v/v as shown in FIG. 3. The eluate corresponding to the washing with aqueous ethanol 80% v/v was dried in rotary evaporator (70° C.), rendering a syrup residue. By addition of chloroform, the residual water in the syrup residue was removed by evaporation, rendering 0.73 g of hydrolyzed extract of saponins containing the Compounds I and II. The total mass

yield of the process described was 0.275 grams of the extract containing Compounds I and II per each gram of VaxSap®.

Example 5: Preparation of Solid and Liquid Media for Cultivation of Microorganisms

[0092] The liquid and solid media employed for the propagation of microorganisms and biotransformation of triterpenic substrates described in the upcoming examples presented in this disclosure is described as follows:

[0093] Liquid media: All liquid media employed were sterilized by autoclaving at 121° C. during 20 min prior to use. When the addition of antibiotics to the media was required, they were prepared in separated stock solutions, filtered through sterile syringe filters (pore size 0.22 micron) and added to the autoclaved broth after cooling.

[0094] Sterile base broth (SBB) was prepared in deionized water (1 L) containing 10.0 g tryptone, 10.0 g sodium chloride and 5 g yeast extract.

[0095] Sabouraud dextrose broth (SDB) was prepared with deionized water (1 L) and contained: 40.0 g glucose, 5.0 g casein and 5.0 g peptone.

[0096] Brain Heart Infusion broth (BHIB) was prepared with deionized water (1 L) and contained: 12.5 g brain infusion solids, 5.0 g beef heart infusion solids, 10.0 g proteose peptone, 5.0 g sodium chloride, 2.0 g glucose and 2.5 g disodium phosphate.

[0097] Solid media: All media employed were sterilized by autoclaving at 121° C. during 20 min prior to distribution in sterile Petri dishes. When the addition of antibiotics to the media was required, they were prepared in separated stock solutions, filtered through sterile syringe filters (pore size 0.22 micron) and added to the autoclaved broth after partial cooling just before to pouring the agar plates.

[0098] Luria Bertani Agar (LBA) was prepared with deionized water (1 L) and contained: 10.0 g tryptone, 5.0 g yeast extract, 10.0 g NaCl and 20.0 g agar-agar.

[0099] Trypticase Soy Agar (TSA) was prepared with deionized water (1 L) and contained: 15.0 g pancreatic digest of casein, 5.0 g peptic digest of soybean meal, 5.0 g sodium chloride and 15.0 g agar-agar.

[0100] Müller Hinton Agar (MHA) was prepared with deionized water (1 L) and contained: 2.0 g beef extract, 17.5 g acid hydrolysate of casein, 1.5 g starch and 17.0 g agar-agar.

[0101] Columbia Agar (CBIA) was prepared with deionized water (1 L) and contained: 10.0 g peptone from casein, 5.0 g peptone from meat, 3.0 g heart extract, 5.0 g extract from yeast, 1.0 g starch, 5.0, sodium chloride and 13.0 g agar-agar.

[0102] De Man, Rogosa and Sharpe Agar (MRSA) was prepared with deionized water (1 L) and contained: 10.0 g enzymatic digest of casein, 10.0 g meat extract, 4.0 g yeast extract, 2.0 g tri-ammonium citrate, 5.0 g sodium acetate, 0.2 g magnesium sulfate heptahydrate, 0.05 g manganese sulphate tetrahydrate, 2.0 g dipotassium hydrogen phosphate, 1.08 g sorbitan mono-oleate, 20.0 g glucose and 12.4 g agar-agar.

[0103] Sabouraud Dextrose Agar (SDA) was prepared with deionized water (1 L) and contained: 40 g glucose, 5.0 g casein, 5.0 g peptone and 15 g agar-agar.

[0104] Brain Heart Infusion Agar (BHIA) was prepared with deionized water (1 L) and contained: 12.5 g brain infusion solids, 5.0 g beef heart infusion solids, 10.0 g

proteose peptone, 5.0 g sodium chloride, 2.0 g glucose, 2.5 g disodium phosphate and 10.0 g agar-agar.

[0105] Potato Dextrose Agar (PDA) was prepared with deionized water (1 L) and contained: 200 g potato infusion, 20.0 g dextrose and 20 g agar-agar.

Example 6: Isolation and Propagation of HNR2 Consortium

[0106] An unpasteurized sample of purified aqueous extract of *Quillaja saponaria* Molina was employed as the source for the isolation of the consortium HNR2. This aqueous extract of *Quillaja saponaria* Molina, stored at room temperature, became spoiled by microbial growth evidenced by a gel like substance floating at the surface of the liquid in one pail of the aqueous extract of *Quillaja saponaria* Molina (FIG. 4). The gel like substance floating at the surface of the aqueous extract of *Quillaja saponaria* Molina was cut into small pieces with a sterile knife cutter (app. 1×1 cm) on a sterile surface and then transferred to Erlenmeyer flasks containing sterile base broth (prepared as described in the Example 5). The flasks containing the inoculated basic broth were incubated overnight at 25° C. on a rotary shaker at 200 rpm. The liquid culture thus obtained (HNR2 consortium) was kept refrigerated at 4° C. and regrown in fresh basic broth under the same conditions described above on weekly basis.

Example 7: Characterization of the Consortium HNR2

[0107] To characterize the HNR2 microbial consortium, 200 microliters aliquots of seed inoculum prepared according to the procedure described in the Example 6 were plated onto Petri dishes containing Luria Bertani Agar (LBA) (prepared as described in the Example 5). The inoculated solid media in Petri dishes were incubated between at 25° C. during 24 to 48 h. Four major types of colonies were grown under the above incubation conditions, hereinafter called colonies C, P, W and Y (FIG. 5).

[0108] Growth on solid media: In order to test the growth of colonies C (RGM 2984), P (RGM 3031), W (RGM 3032) and Y (RGM 3021) in different solid media, they were plated on the following solid media described in the Example 5: Trypticase soy agar (TSA), Muller-Hinton agar (MHA), Columbia agar (CBIA), De Man, Rogosa and Sharpe agar (MRSA), Sabouraud dextrose agar (SDA), brain heart infusion agar (BHIA), and potato dextrose agar (PDA). The inoculated solid media in Petri dishes were incubated between at 25° C. during 24 to 48 h. The growth behavior of these colonies is summarized in the Tables 2 and 3.

TABLE 2

Growth of colonies C, P, W and Y in solid media after 24 hours at 25° C. The growth was qualitatively classified as significant (++), slight (+) and non-observable (-).								
Colony	LBA	TSA	MHA	CBIA	MRSA	SDA	BHIA	PDA
W	+	++	+	+	-	-	+	-
Y	+	+	+	+	-	-	+	-
P	+	+	+	+	-	-	+	+
C	++	++	++	++	++	++	++	++

TABLE 3

Growth of colonies C, P, W and Y in solid media after 48 hours at 25° C. The growth was qualitatively classified as significant (++), slight (+) and non-observable (-).								
Colony	LBA	TSA	MHA	CBIA	MRSA	SDA	BHIA	PDA
W	+	++	+	++	-	-	++	-
Y	++	++	+	++	-	-	++	-
P	+	+	++	++	++	+	+	++
C	++	++	++	++	++	++	++	++

[0109] Growth on solid media supplemented with antibiotics: In order to test the resistance to chloramphenicol, ampicillin, kanamycin and streptomycin of the colonies C, P W and Y during the growth in solid media, they were plated onto Petri dishes containing Luria Bertani Agar (colonies W and Y) and potato dextrose agar (colonies C and P) supplemented with antibiotics as described in the Tables 4 and 5, and incubated 48 hours at 25° C.

TABLE 4

Growth of colonies C and P on solid media after 48 hours at 25° C.		
Solid media	Colony C	Colony P
PDA + chloramphenicol 20 microg/ml	Resistant	Resistant
PDA + ampicillin 20 microg/ml	Resistant	Resistant
PDA + kanamycin 20 microg/ml	Resistant	Resistant
PDA + streptomycin 20 microg/ml	Resistant	Resistant

TABLE 5

Growth of colonies W and Y on solid media after 48 hours at 25° C.		
Solid media	Colony W	Colony Y
LBA + chloramphenicol 20 microg/ml	Sensitive	Sensitive
LBA + ampicillin 20 microg/ml	Resistant	Resistant
LBA + kanamycin 20 microg/ml	Sensitive	Resistant
LBA + streptomycin 20 microg/ml	Resistant	Sensitive

[0110] Morphology and biochemical characterization: The morphology of the colonies C, P W and Y was determined after growth on Petri dishes containing Luria Bertani Agar (colonies W and Y) and potato dextrose agar (colonies C and

Bergey's Manual® of Systematic Bacteriology. The results are summarized in the Table 6.

TABLE 6

Summary of the characterization of the characterization of the colonies C, P (grown on PDA medium), and W, Y (grown on LDA medium) after 48 hours at 25° C.				
Parameter	Colony C	Colony P	Colony W	Colony Y
Chromogenesis	White	Pink	White	Yellow
Shape of the colony	Rhizoid	Circular	Filamentous	Circular
Surface of the colony	Umbonate	Convex	Flat	Convex
Edge of the colony	Filiform	Entire	Undulate	Entire
Opacity	Opaque	Opaque	Translucent	Opaque
Gram staining/shape observed in the microscope	Not determined	Not determined	Negative/bacillus	Negative/coccus
Lactophenol cotton blue staining/shape observed in the microscope	Positive staining/filamentous	Positive staining/spherical	Not determined	Not determined
Catalase	Not determined	Not determined	Negative	Positive
Amylase	Not determined	Not determined	Positive	Negative
Gelatinase	Not determined	Not determined	Negative	Negative

[0111] Characterization by ribosomal sequence analysis: In order to complete the characterization of the colonies C, P, W and Y the ribosomal sequence analysis was performed as follows:

[0112] Microbial propagation: A single colony of C, P W and Y strains was inoculated in 4 ml of the media listed in the Table 7, and grown overnight at 25° C. on a rotary shaker at 200 rpm.

TABLE 7

Liquid culture media employed for the propagation of the colonies C, P, W and Y as required for their characterization by ribosomal sequence analysis.						
Strain	Broth	Glucose (added as sterile stock solution to the autoclaved broth)	Ampicillin (added as sterile stock solution to the autoclaved broth)	Chloramphenicol (added as sterile stock solution to the autoclaved broth)	Kanamycin (added as sterile stock solution to the autoclaved broth)	Streptomycin (added as sterile stock solution to the autoclaved broth)
W	SBB	10 g/L	—	—	—	20 mg/L
Y	SBB	10 g/L	—	—	20 mg/L	—
P	SDB	10 g/L	—	20 mg/L	—	—
C	SBB	10 g/L	50 mg/L	20 mg/L	—	—

P). Samples of each type of colony were used for morphology and biochemical studies following the procedures in

[0113] Total DNA isolation: Each culture was centrifuged 5 min at 20,000×g. Total genomic PG 37T DNA was isolated

from the centrifuged pellets using two commercial kits: E.Z.N.A.® Bacterial DNA Kit (Omega Bio-Tek) for colonies W and Y, and DNeasy Plant Mini Kit (QIAGEN) for colonies C and P. In each case, the DNA was isolated following the instruction of the manufacturer. The DNA of each colony was recovered in 100 µL of buffer TE (Tris 10 mM, EDTA 1 mM, pH 8.0). The purity and yield of the DNA preparations were determined by measurement of the absorbance at 260 and 280 nm of each DNA solution. The DNA solutions were stored at −20° C.

[0114] Primers for amplification of ribosomal sequences from total DNA by polymerase chain reaction (PCR): Table 8 summarizes the nucleotide sequences of the DNA primers employed for the amplification of ribosomal sequences from total DNA isolated from colonies C, P, W and Y.

TABLE 8

Nucleotide sequences of the DNA primers employed for the PCR amplification of ribosomal sequences from total DNA.				
Pair.	name.	Nucleotide Primer sequence (5'→3').	Target sequence	Expected amplicon size [base pairs]
PP1	27F	AGAGTTTGATCMTGG CTCAG (SEQ ID No.: 1)	Bacterial ribosomal sequence	1450
	R1492	TACGGYTACCTTGTT ACGACTT (SEQ ID No.: 2)	16S rDNA (*)	
PP2	63F	CAGGCCTAACACATG CAAGTC (SEQ ID No.: 3)	Bacterial ribosomal sequence	1500
	M1387R	GGGCGGWTGTACAA GRC (SEQ ID No.: 4)	16S rDNA (*)	
PP3	NL-1	GCATATCAATAAGCG GAGGAAAAG (SEQ ID No.: 5)	Fungal D1/D2 region of	680
	NL-4	GGTCCGTGTTTCAAG ACGG (SEQ ID No.: 6)	the ribosomal sequence	
			28S rDNA (**)	
PP4	ITS1	TCCGTAGGTGAACCT GCGG (SEQ ID No.: 7)	Fungal ribosomal sequence	620

TABLE 8-continued

Nucleotide sequences of the DNA primers employed for the PCR amplification of ribosomal sequences from total DNA.			
Pair.	name.	Nucleotide Primer sequence (5'→3').	Target sequence
	ITS4	TCCTCCGCTTATTGAT ATGC (SEQ ID No.: 8)	5S rDNA (***)

(*) Fredriksson, N. J., Hermansson, M., & Willén, B. M. 2013. "The choice of PCR primers has great impact on assessments of bacterial community diversity and dynamics in a wastewater treatment plant". PLoS One. 8(10): e76431.

(**) Fernández-Bodega, M. A., Mauriz, E., Gómez, A., & Martin, J. F. 2009. "Proteolytic activity, mycotoxins and andrastin A in *Penicillium roqueforti* strains isolated from Cabrales, Valdeón and Bejes-Tresviso local varieties of blue-veined cheeses". International journal of food microbiology. 136(1): 18-25.

(***) Gardes M, Bruns TD. 1993. "ITS primers with enhanced specificity for basidiomycetes - application to the identification of mycorrhizae and rusts". Molecular Ecology. 2: 113-118.

[0115] FIGS. 6A, 6B, 6C and 6D show the results of the agarose electrophoresis analysis of the PCR products obtained by amplification of the genomic DNA of the colonies C, P, W and Y, with the primer pairs PP1 (SEQ ID Nos.: 1 and 2), PP2 (SEQ ID Nos.: 3 and 4), PP3 (SEQ ID Nos.: 5 and 6) and PP4 (SEQ ID Nos.: 7 and 8), respectively. The results shown in FIGS. 6A and 6B confirm that colonies W and Y, are prokaryotic bacterial cells. The results shown in FIGS. 6C and 6D confirm that colonies C and P, are fungal eukaryotic cells.

[0116] The PCR products described previously were sequenced (Macrogen, Seoul, Korea) (Sequences SEQ ID No. 9 a 23. DNA sequences of the PCR product obtained from genomic DNA of colony C with primer pair PP3 (SEQ ID Nos.: 5 and 6). DNA sequences of the PCR product obtained from genomic DNA of colony C with primer pair PP4 (SEQ ID Nos.: 7 and 8). DNA sequences of the PCR product obtained from genomic DNA of colony P with primer pair PP3 (SEQ ID Nos.: 5 and 6). DNA sequences of the PCR is product obtained from genomic DNA of colony P with primer pair PP4 (SEQ ID Nos.: 7 and 8). DNA sequences of the PCR product obtained from genomic DNA of colony W with primer pair PP1 (SEQ ID Nos.: 1 and 2). DNA sequences of the PCR product obtained from genomic DNA of colony W with primer pair PP2 (SEQ ID Nos.: 3 and 4). DNA sequences of the PCR product obtained from genomic DNA of colony Y with primer pair PP1 (SEQ ID Nos.: 1 and 2). DNA sequences of the PCR product obtained from genomic DNA of colony Y with primer pair PP2. (SEQ ID Nos.: 3 y and 4)). The DNA sequences were edited and aligned against published ribosomal sequences of prokaryotic and eukaryotic microorganisms employing the software NCBI BLAST (<http://www.ncbi.nlm.gov/>) and ClustalW (multiple sequence alignment in BioEdit v7.0.9 (<http://www.mbio.nesu.edu/BioEdit/bioedit.html>)). The list of the most probable microorganisms corresponding to each type colony under study is summarized in the Table 9. According to the results, colonies C, P, W and Y most likely correspond to *Mucor* spp, *Rhodotorula* spp, *Paenibacillus* spp and *Microbacterium* spp, respectively.

TABLE 9

List of the most probable microorganisms corresponding to the colonies C, P, W and Y, as determined by alignment of ribosomal sequences employing CLUSTAL W.					
Colony	Probable type of microorganism according to sequence alignment	Sequence homology (%)			
		Primer pair PP1 (SEQ ID Nos.: 1 and 2)	Primer pair PP2 (SEQ ID Nos.: 3 and 4)	Primer pair PP3 (SEQ ID Nos.: 5 and 6)	Primer pair PP4 (SEQ ID Nos.: 7 and 8)
C	<i>Mucor racemosus</i> strain ATCC 42647	—	—	100%	99%
C	<i>Mucor racemosus</i> strain X1650	—	—	100%	99%
P	<i>Rhodotorula mucilaginosa</i>	—	—	95%	100%
P	<i>Rhodotorula mucilaginosa</i>	—	—	95%	99%
W	<i>Paenibacillus amylolyticus</i>	99%	80%	—	—
W	<i>Paenibacillus tylopili</i>	99%	80%	—	—
W	<i>Paenibacillus tundrae</i>	99%	80%	—	—
Y	<i>Microbacterium phyllosphaerae</i>	99%	89%	—	—
Y	<i>Microbacterium hydrocarbonoxydans</i>	99%	84%	—	—
Y	<i>Microbacterium foliomm</i>	99%	84%	—	—

Example 8: Production of SPH03 from an Extract Comprising Compounds I and II

[0117] To produce the seed inoculum, an aliquot of a refrigerated culture HNR2 as defined above, was inoculated on 5 ml of sterile base broth prepared according to the recipe described in the Example 5. The cells were grown overnight at 25° C. on a rotary shaker at 200 rpm, pelleted by centrifugation and washed with 5 ml of sterile base broth. Washed cells were resuspended on 5 ml of sterile base broth rendering the seed inoculum.

[0118] To perform the biotransformation of the hydrolyzed extract of saponins of *Quillaja saponaria* Molina, shake flasks (250 ml) containing 45 ml of base broth were sterilized at 1210° C. for 20 min. After sterilization, 100 microliters of filter sterilized glucose (200 g/L) and 2.0 ml of filter sterilized an extract comprising Compounds I and II (24 g/L) were blended into the sterilized base broth. The flasks were cooled to 25° C. and inoculated with 1.5 ml of seed inoculum prepared according to the procedure described in the Example 6. The flasks were then incubated at 25° C. on a rotary shaker (200 rpm) during one week and sampled at various intervals. One (1) ml samples were

removed and centrifuged to remove cells. Progress of the biotransformation was followed by UHPLC-PDA and TLC analysis of the supernatants of each sample, according to the procedure described in the Examples 1 and 3. The UHPLC analysis of the supernatant of the culture shows no trace of Compounds I and II, and the rising of a new peak with increased retention time, hereinafter denominated as SPH03 is shown in FIG. 7, confirms the biotransformation of the extract containing the Compounds I and II by HNR2 culture. Similarly, TLC analysis of culture supernatants, also showed the disappearance of the Compounds I and II, and the appearance of SPH03 as shown in FIG. 8.

Example 9: Effect of Broth Composition on the Production of SPH03 from an Extract Containing Compounds I and II

[0119] The impact of the broth formulation on SPH03 production from partially hydrolyzed saponins (Compounds I and II) during the biotransformation process was assessed on a range of liquid broths based on SBB and BHIB broths (Example 5) supplemented with some carbohydrates as described in the Table 10. SPH03 production in the cultures is formulated according to the Table 10 was determined by TLC chromatography (FIG. 9).

TABLE 10

Effect of the broth formulation on the production of SPH03 from an extract containing Compounds I and II, catalyzed by the HNR2 consortium: Results after 120 h of incubation.						
TLC lane FIG. 10	Liquid broth composition	Inoculum HNR2	Extract containing Compounds I and II	Production of SPH03?	Time for complete conversion [h]	
1	SBB	+	1 g/L	+	48	
2	SBB + glucose 1.0 g/L	+	1 g/L	-	-	
3	SBB + sucrose 1.0 g/L	+	1 g/L	-	-	
4	SBB + lactose 1.0 g/L	+	1 g/L	-	-	
5	SBB	-	1 g/L	-	-	
6	BHIB	+	1 g/L	-	-	
7	BHIB + glucose 0.4 g/L	+	1 g/L	-	-	

TABLE 10-continued

Effect of the broth formulation on the production of SPH03 from an extract containing Compounds I and II, catalyzed by the HNR2 consortium: Results after 120 h of incubation.					
TLC lane FIG. 10	Liquid broth composition	Inoculum HNR2	Extract containing Compounds I and II	Production of SPH03?	Time for complete conversion [h]
8	SBB + sucrose 0.4 g/L	+	1 g/L	–	–
9, 10	SBB + xylan 1.0 g/L	+	1 g/L	+	36
11	SBB + lactose 0.4 g/L	+	1 g/L	+	96
12	SBB + glucose 0.4 g/L	+	1 g/L	+	48

[0120] According to the results shown in Table 10, BHIB broth as well as the addition of sucrose 0.4 g/L to SBB inhibits the production of SPH03. The addition of glucose and lactose 1.0 g/L inhibits SPH3 production. The quickest conversion of the Compounds I and II to SPH03 is achieved in SBB broth supplemented with xylan 1 g/L. The impact of the substrate concentration for the biotransformation (extract containing Compounds I and II) and xylan on SPH3 production during the biotransformation process, was assessed on a range of SBB based broths formulated as described in the Table 11. The conversion of extract containing Compounds I and II in SPH03 in the cultures formulated according to the Table 11 was determined by UHPLC chromatography of samples taken at regular time intervals (FIG. 10).

TABLE 11

Effect xylan and Compounds I and II concentrations on the production of SPH03 from an extract containing Compounds I and II, catalyzed by HNR2 consortium.				
Extract containing Compounds I and II [g/L]	Xylan [g/L]	Conversion to SPH03 [conversion %]	Time for maximum conversion [h]	Productivity of SPH03 at the time for maximum conversion [mg/L/h]
1.5 g/L	2.5 g/L	100%	32	47
3.0 g/L	2.5 g/L	100%	52	58
4.5 g/L	2.5 g/L	24%	73	15
1.5 g/L	3.5 g/L	100%	71	21
1.5 g/L	5.0 g/L	0%	ND	0

[0121] According to the results shown in Table 11, SBB broth formulated with 3.0 g/L of extract containing Compounds I and II, and 2.5 g/L xylan allows to maximize the SPH03 production in batch cultures of HNR2 consortium.

Example 10: Purification of SPH03 from the Biotransformation Broth

[0122] As a representative example of SPH03 recovery we provide a detailed description of the purification procedure as applied to the recovery of the biotransformation product from the cultivation broth of HNR2 grown on SBB broth formulated with 1.0 g/L of extract containing Compounds I and II, and 2.5 g/L xylan. The following description is not limited to the purification of SPH03 from the cultivation broth obtained in the present examples as well as the

Examples 8 and 9, but also applicable to the processing of biotransformation broths obtained in other conditions. The cultivation broth (300 ml) was centrifuged at 3220×g for 10 min in order to remove the microbial cells. The supernatant is was recovered, and the pH adjusted to 5.5 by addition of formic acid 1 M and 5% v/v of acetonitrile. The resulting solution was concentrated to 1/5 of the initial volume by evaporation at reduced pressure at 36° C. rendering a concentrated precursor of SPH03. Aliquots of the concentrated precursor of SPH03 (1.4 ml) were injected in a semipreparative reverse phase HPLC column (inner diameter 21.2 mm; length 250 mm; particle size 7 µm) and eluted in gradient mode at constant flow rate (30 ml/min) employing the following mobile phases: Solvent A: Formic acid 0.15% v/v in water, and; Solvent B: Formic acid 0.15% v/v in acetonitrile. The solvent gradient employed is described as follows: Step 1: 0-0.4 min 5% of Solvent B; Step 2: 0.4-24 min from 5% to 40% of Solvent B; Step 3: 24-35 min 40% of Solvent B; Step 4, and; 35-47 min from 40% to 5% of Solvent B.

[0123] The saponin profile of the collected fractions is then determined by reverse phase UHPLC according to the conditions described in the Example 1. Fractions containing the target saponin were selected for further purification. The selected fractions of 45 preparative runs were pooled and solvent was removed by evaporation at reduced pressure at 35° C. The residual aqueous solution was lyophilized, rendering the purified SPH03.

[0124] At the end of the drying procedure 88 mg of purified SPH03 were recovered. The total mass yield of the process described was 0.29 g of SPH03 per g of an extract containing Compounds I and II. FIG. 12 provides a representative RP UHPLC chromatogram of the SPH03 product purified by preparative RP HPLC.

Example 11: Identification of Triterpenic Components in SPH03

[0125] The components in SPH03 were identified by HPLC-ESI-TOF-MS as described in the is Example 2. In the SPH03 fraction, four co-eluting compounds were found, herein referred as Compounds III, IV, V and VI. The mass of the pseudo-molecular ions $[M+NH_4]^+$ was detected in the mass spectra of Compounds III, IV, V and VI, as shown in FIGS. 13, 14, 15 and 16, respectively. The molecular mass of the ammonium adducts and acid forms of Compounds HI, IV, V and VI, are summarized in the Table 12.

TABLE 12

Molecular mass of the ammonium adducts and acid forms of Compounds III, IV, V and VI detected by HPLC-ESI-TOF-MS.		
Compound	Molecular mass of the ammonium adducts. [Dalton]	Molecular mass of the acid form. [Dalton]
III	842	825
IV	812	795
V	826	809
VI	680	663

[0126] Compound III has been referred previously in the literature as a minor saponin in the extracts of the bark of *Quillaja saponaria* Molina, and as degradation product during acid hydrolysis of the saponins of quillaja Guo et al., *Phytochemistry*. 48: 175-180 (1998). Compound VI has been described by Labriola & Deulofeu (*Experientia*. 25: 124-125 (1969)) as a product of acid hydrolysis of *Quillaja saponins* with hot acid. The Compounds IV and V have not been previously detected in quillaja extracts.

Example 12: Hemolytic Activity of SPH03

[0127] The hemolytic activity of SPH03, an extract containing Compounds I and II and purified *Quillaja saponins* "VetSap" (Desert King International) was determined by the following method: Serial dilutions of each sample in PBS buffer were made in Eppendorf tubes (250 microL). All samples were tested in duplicate. Chicken red blood cells (CRBC) were washed three times with PBS, and were diluted to 2% with PBS. CRBC (250 microL) were added to each tube. A mixture of *Quillaja saponins* "VetSap" 500 mg/L (250 microL) and CRBC (250 microL) was prepared as positive control. A mixture of PBS (250 microL) and CRBC (250 microL) was prepared as negative control. The contents of all tubes were gently mixed by inversion and incubated at 37° C. by 1 h. After incubation, the tubes were spun at 18,620xg to sediment non hemolyzed cells. 100 microL of the supernatant from each tube was transferred to separate wells in a flat bottom microtiter plate. Absorbance was determined at 540 nm against water blank with an Epoch microplate spectrophotometer. (BioTek Instruments Inc.). Hemolysis increased the absorbance at 540 nm due to release of hemoglobin from the lysed cells. The value of the absorbance of the negative control was subtracted to all remaining absorbance values corresponding to samples and positive control, rendering the Δ Abs values. The hemolysis percentage was determined dividing each Δ Abs by the corresponding Δ Abs of the positive control (FIGS. 16A and 16B). Significant differences in hemolysis were observed between adjuvants the tested samples. *Quillaja saponins* "VetSap" caused partial hemolysis at concentrations as low as 3.9 mg/L; complete hemolysis was observed with concentrations in the range 7.8-62.5 mg/L (FIG. 16A). Negligible hemolysis (less than 5%) was observed with SPH03 and the extract containing Compounds I and II in the same range of concentrations tested (3.9-62.5 mg/L; FIGS. 16A and 16B).

Example 13: Stability of SPH03 at Physiological pH

[0128] Fresh solutions 0.5 g/L of SPH03 and high purity *Quillaja saponins* were prepared in Mc Ilvine buffer (pH

7.4) incubated at 37° C. during 88 h. The stability of each solution was determined by UHPLC chromatography of samples taken at 0, 2 and 88 hours. The results confirm the SPH03 stability at physiological pH. Under the same conditions, *Quillaja saponins* were partially degraded (FIG. 17).

Example 14: Cytotoxic activity of SPH03

[0129] A solution (1 mM) of SPH03 and the extract containing Compounds I and II were prepared on PBS buffer. The cytotoxic activity of SPH03 fraction and the extract containing Compounds I and II were tested on two types of tumor cells of human gastric cancer already available at the American Type Culture Collection (ATCC): SNU1 (gastric carcinoma) and AGS (gastric adenoma), a human gastric epithelial cell line GES-1 used as normal cell control. All the cell lines were grown on RPMI 1640 broth or DEMEN supplemented with 10% FBS and 1% penicillin/streptomycin, under CO₂ rich atmosphere (5%) at 37° C. Grown cells were transferred to fresh broth and propagated until 80% density. Then, the cell culture containing 5,000 cells was transferred to 96 well plates for further growth during 24 h. After growth, the stock solutions of SPH03 and the extract containing Compounds I and II were added to the wells in order to test the cytotoxic effect of each solution in a range of concentrations covering 0 to 150 μ M and is 0 to 100 μ M, respectively. Cell suspensions were further incubated during 48 and 72 h at 37° C. under CO₂ atmosphere (5%). Cell viability was determined spectrophotometrically at 490 nm using the reagent CellTiter 96@Aqueous One Solution Proliferation Assay (MTS) supplied by Promega (Madison, WI, USA). Each condition was tested in triplicate and three independent assays. To obtain IC₅₀ value to SPH03, dose-response curve was constructed to SNU1 and AGS cell lines using GraphPad Prism 8 (GraphPad Software, San Diego, CA, USA).

[0130] The results of IC₅₀ obtained are shown in Table 13. After incubation during 72 h, the extract containing Compounds I and II had not severe impact on cell viability of AGS and GES-1 cell lines in all the ranges of concentration assayed (FIG. 18a). However, a slight effect was observed on the viability of SNU1 cells in the range 20 to 30 μ M. On the other hand, the SPH03 fraction was able to abolish above a 40% of the cell viability of tumoral SNU1 cells at 50 μ M (FIG. 18b). When the concentration of SPH03 fraction was increased to 75 μ M, the viability of SNU1 and AGS cells was almost abolished (9%), whereas 51% of GES-1 cells remained viable ($p < 0.0001$) (FIG. 18b). The latter results show a significant difference of the cytotoxic effect of SPH03 on normal cell vs cancer cell lines.

TABLE 13

In vitro anti-proliferative activity of an extract containing Compounds I and II and SPH03 on gastric cancer cell lines (SNU1 and AGS) and normal human gastric cells (GES-1). The IC ₅₀ values represent the average of at least 3 independent experiments.			
Tested sample	IC ₅₀ (μ M)		
	SNU1 cells	AGS cells	GES-1 cells
SPH03	47.72	59.94	69.09
Extract containing Compounds I and II	>100	>100	>100

[0131] To confirm the pro-apoptotic effect of SPH03, the treated cells were assayed by Muse Annexin V as well as a dead cell kit (Merck, Millipore, USA), according to the procedures recommended by each manufacturer. To this end, $2 \cdot 10^5$ cells (AGS and SNU1) were seeded in 6-well plates and treated with different concentrations of SPH03 (35, 50 and 75 μM) by 48 h. Cisplatin drug (50 μM) was employed as positive control of apoptosis. Cultured AGS and SNU cells not supplemented with SPH03 were employed as negative control. After incubation, the cells were washed with 1 mL PBS and 100 μL of Muse™ Annexin V and 100 μL of Dead Cell reagent were added. Then, the cells were incubated at room temperature during 20 min in the darkness. The apoptotic effect was measured by using Muse cell analyzer and Muse analysis software (Merck-Merck Millipore), where the cells were classified into four groups: live, early apoptotic, late apoptotic and dead or necrosis. The percentage of AGS cells undergoing apoptotic cell death (early and late stages) increased to 37.45% and 42.8% after exposure to 50 and 70 μM of SPH03, respectively (FIG. 19). Untreated AGS cells cultured as negative control were 97.34% viable (FIG. 19). In the case of SNU1 the percentage of cells undergoing apoptotic cell death (early and late stages) increased to a 40.3 and 64.5%, after exposure to 50 μM and 70 μM of SPH03, respectively (FIG. 20). Untreated SNU1 cells cultured as negative control were 90.29% viable (FIG. 21).

Example 15: Inhibition of Biofilm Formation by
Streptococcus mutans UA159 and *Enterococcus faecalis* ATCC 29212 Grown in Broth
Supplemented with SPH03

[0132] To measure the impact of SPH03 on the formation of biofilms by *Streptococcus mutans* UA159 and *Enterococcus faecalis* ATCC 29212, an assay based on the binding of crystal violet dye to biofilms was applied. *Streptococcus mutans* UA159 and *Enterococcus faecalis* ATCC 29212 were inoculated on BHIB broth (Example 5) and cultured by 18 h at 37° C. in a shaking incubator. Once the optical density at 600 nm (OD600) of each culture reached 0.1 AU, 10 microL aliquots were transferred to in 96-well plates containing 140 microL of BHIB broth supplemented with glucose (10 g/L) and a range of SPH03 covering from 0 to 3.2 g/L. The plate was incubated at 37° C. during 24 h. Then, the culture was removed of each well in the plate, followed by gentle washing with PBS buffer in order to remove free planktonic cells. The plate containing the bacterial biofilms was incubated 1 h at 60° C., followed by the addition of 200 microL of aqueous 4% crystal violet solution to each well, followed by incubation at room temperature during 15 min. The plate was washed and dried at 60° C. during 15 min. To dissolve the remaining crystal violet attached to biofilms, 200 microL of acetic acid 0.33% were added to each well, and the absorbance measured at 570 nm. The resultant absorbance measured in each condition is proportional to the amount of biofilm in the corresponding well. The results show a significant effect of SPH03 at concentrations higher than 1.9 g/L when compared with control cells (FIG. 21).

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ctttggaaca gggcatcata gaggggtgaga atcccgctct tgggtctaagt agttgtctat    180
tgtgatatgt tttcaaagag tcagggtgtt tgggaatgca gcctaaattg ggtggtaaat    240
ctcacctaaa gctaaatatt tgcgagagac cgatagcgaa caagtaccgt gagggaaaga    300
tgaaaagaac tttgaaaaga gagttaaaca gtatgtgaaa ttgttaaaag ggaaccgttt    360
ggagccagat tggcttgatt gtaatcaacc tagaattcgt tttgggtgca cttgcagtct    420
atgctcgcca acgacagttt gattttggagg aaaaaattaa taggaatgtg gcctttcgag    480
gtgttatagc ctattatcat actctggatt ggactgagga acgcagcgaa tgcctttagg    540
caagattgct gggcgctttc gctaataaat gttagaattt ctgcttcggg tgggtgctaat    600
gtttaaagga ggaacccgtt tagtatattt tttattcgtc taggtgtgtg gcttaatgac    660
tctaaatgac ccgtctaaaa ccggggga                                           687

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<210> SEQ ID NO 11
<211> LENGTH: 1000
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP4/ITS1, reverse sequence

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<400> SEQUENCE: 11

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gggctaaata tcatatcttg gcttgccat tattatctat ttactgtgaa ctgtattatt    60
atttgacatt tgagggatgt tccaatgtta taaggataga cattggaaat gttaaccgag    120
tcataatcag gtttaggcct ggtatcctat tattatttac caaatgaatt cagaattaat    180
attgtaacat agacctaaaa aatctataaa acaactttta acaacggatc tcttggttct    240
cgcacgatg aagaacgtag caaagtgcga taactagtgt gaattgcata ttcagtgaat    300
catcgagtct ttgaacgcaa cttgcgtca ttggtattcc aatgagcacg cctgtttcag    360
tatcaaaaca aacctcttat tcaacttttg ttgtatagga ttattggggg cctctcgatc    420
tgtatagatc ttgaaatccc tgaaatttac taaggcctga acttgtttaa atgctggaac    480

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tttttttaa tataaaggaa agctcttgta attgactttg atggggcctc ccaaataaat 540
cttttttaaa tttgatctga aatcaggcgg gattaccgcg tgaacttaag catatctaaa 600
cgagaaaaaa aaaaaatcc tttaaataat cattaatctg ggctggctct taataatcta 660
tttacggtga aatggtataa ttatttgaca ttgagggga tgtccaatg tttataaggg 720
atagactttg gaaaatgtta accgagtcga aatcgggttt aagcctgggt atccaattaa 780
ttattttaca aaggaaatc aaattaaaat tgtttacaaa gaactaaaa aatctattaa 840
aaaaacttta acaacggaaa ctttgggttt ctgcaccctt taaaaaaagt aacaaaatgg 900
caataactag tggggaaatt gcaaaattcc ctggaaatca cccggttctt ttgaaagcca 960
cctgggcctc aatgggggtt tccccatgaa gcaagccccg 1000

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<210> SEQ ID NO 12
<211> LENGTH: 850
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP4/ITS2, forward sequence

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<400> SEQUENCE: 12

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tgatctcgct tccgcctgat ttcagatcaa tttaaaaaag atttatttgg gagggcccat 60
caaagtcaat ttacaagagc tttcctttat attaaaaaaa agttcaggca tttaaacaag 120
ttcaggcctt agtaaatttc agggatttca agatctatac agatcgagag gcccccaata 180
atcctataca acaaaagtgt aatagagggt ttgttttgat actgaaacag gcgtgctcat 240
tggaatacca atgagcgcaa gttgcgttca aagactcgat gattcaactga atatgcaatt 300
cacactagtt atcgcacttt gctacgttct tcatcgatgc gagaaccaag agatccgttg 360
ttaaaagtgt ttttatagat tttttaggtc tatgttacia tattaattct gaattcattt 420
ggtaataaat aataggatac caggcctaaa cctgattatg actcgggtta catttccaat 480
gtctatcctt ataacattgg aacatccctc aaatgtcaaa taataatata gttcacagta 540
aatagataat aatggactag ccaagattat tgattattta atgaccttc cgcactctcc 600
caccggaaaa aagaaaaaat cctttaaaaa aaccaataaa ccggggccgg gcccttaaaa 660
atttttttcc cggtgaaatg gtaaaaaaaa atttgaaatt tggaaggga gggttccaat 720
gtttataaag gaaaagaact ttgggaaaat gtttaaccga gttcaaaaac cgggttttag 780
ggccggggta tcccatataa aaattttacc aaaggaaaat ccccaaaata aaaatttgg 840
gaacaaaaga 850

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<210> SEQ ID NO 13
<211> LENGTH: 673
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP3/NL-1, reverse sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (577)..(577)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (591)..(592)
<223> OTHER INFORMATION: n is a, c, g, or t

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<400> SEQUENCE: 13

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catcctaggt gtgagaggtc gacccccgcc taaggcacac tgggttcctc agtcccccaa    60
gatgtatccc gcagagagct ataacacagc cgaaactgct accttctctc taccattatc    120
catcccggaa aactgatgct ggctcgcaaa ccgatggctc ggcaagcaag tctgacttcc    180
agcgattccc ttccaacaat ttcacgtact gttaactctc tttccaaagt gcttttctac    240
tttcctctac ggtacttggt cgctatcggt ctcccgccaa tatttagctt tagatggaat    300
ttaccaccca atttgagctg cattcccaaa caactcgaaa cttcgaaaat gtatcacaaa    360
tggctggggc tccgcaccat atacgggggt ctcttacta tgccgtata ttccaaaaaa    420
cttgtgtgcg gtccaacgcg gaaaacattt ctagagatta caactgggag accgaaagag    480
ccagattata aatttgagct cgctccgctc cctcgcgcgc tacgaagggg aatccatggg    540
cgtttcaatt cccccctga aaaaaagca aaaaaantcg ggaaacccct nngtatactc    600
tggagcagtc tcatgtacta cgacatacag aactctcttg cagtatgtct actccctcgt    660
ggggaaaaag gac                                                    673

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<210> SEQ ID NO 14
<211> LENGTH: 687
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP3/NL-4, reverse sequence

<400> SEQUENCE: 14

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tggggcattt ccctagtagc gcgagtgaga ggaagagctc aaagtggaa cctgtttggc    60
ttagctaaac cggattgtaa actgtagaaa cattttccag atacactaga caaaaaagtc    120
ctttggaaca gggcatcata gagggtgaga atcccgtctt tggctaaagt agttgtctat    180
tgtgatatgt tttcaaaagc tcagggtggt tgggaatgca gcctaaattg ggtggtaaat    240
ctcacctaaa gctaaatatt tgcgagagac cgatagcgaa caagtaccgt gagggaaaga    300
tgaaaagaac tttgaaaaga gagttaaaca gtatgtgaaa ttgttaaaag ggaaccgttt    360
ggagccagat tggcttgatt gtaatcaacc tagaattcgt tttgggtgca cttgcagtct    420
atgcctgcc aacgacagttt gatttggagg aaaaaattaa taggaatgtg gcctttcgag    480
gtgttatagc ctattatcat actctggatt ggactgagga acgcagcgaa tgcctttagg    540
caagattgct gggcgcttct gctaataaat gttagaattt ctgcttcggg tgggtgctaat    600
gtttaaagga ggaacccgtt tagtatatct tttattcgtc taggttggtg gcttaatgac    660
tctaaatgac ccgtctaaaa ccgggga

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<210> SEQ ID NO 15
<211> LENGTH: 1260
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP4/ITS1, reverse sequence

<400> SEQUENCE: 15

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acggaattgg aaaatagtaa gctcacattc tcatcaaaag tgtttctggg gaagttctga    60
ccctgtgcac ttgtttggga tagtaactct cgcaagagag cgaactccta ttcacttata    120
aacacaaagt ctatgaatgt attaaatttt ataacaaaat aaaactttca acaacggatc    180
tcttggctct cgcacgatg aagaacgcag cgaaatgcga taagtaatgt gaattgcaga    240

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attcagtga tcatogaatc tttgaacgca ccttgcgctc catggtattc cgtggagcat 300
gcctgtttga gtgtcatgaa tacttcaacc ctectcttcc ttaatgattg aagaggtgtt 360
tgggtttctga gcgctgtctg cctttacggt ctagctcgtt cgtaatgcat tagcatccgc 420
aatcgaactt cggattgact tggcgtaata gactattcgc tgaggaattc tagtcttcgg 480
actagagccg gggtgggtta aaggaagctt ctaatcagaa tgtctacatt ttaagattag 540
atctcaaatc aggtaggact acccgctgaa cttaagcata tctaaacccg aaaggaaagg 600
aatcattagg ggaatatagg aacgtccact taacttgga gtcggaactc tcactttcta 660
accctgtgca cttgttttgg gatagttaac tcttgcaaga gagcgaactc ctattcactt 720
ataaacacaa agtctatgaa aggtattaaa atttttataa caaaattaaa attttcacia 780
cggaactttt ggggctctgg catcgaataa aaaaagcaa cgaaaatgcg gattagttaa 840
atggtggaaa tggcagaaaa ttcattgaa atcacgaaa ccttttgaac gcacctggc 900
gtccccgggg taattccctg ggaacaagg cccgttttgg aagttgttca ggaaaaaac 960
attcaacctt cccctttttt cttaaaaagg aattggaaaa aaaaggtttt tttttgggtt 1020
ttttcctgaa agcccttcc tgggcccctt ttaaggggtt ttaaaccttg tttttgtta 1080
aaagggccaa attaaaaaa ccccccaat tccaaaaaat ttttcgaaa aattggaaac 1140
ctttggggcc cctttaaat taaaaaaa caaaatttcc ccccttgaaa aggggaaaaa 1200
atttttcttt ggggtttccc ttccgcggg aaacattaaa aaaaagccc cccggggggt 1260

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<210> SEQ ID NO 16
<211> LENGTH: 617
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP4/ITS2, forward sequence

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<400> SEQUENCE: 16

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aatgggaatt taggaggtcc acttaacttg tcagtcgaa ctctcttcc gctcctgtga 60
tctgttttaa ccaaccccg ctctagtcg aagactagaa tctctcagc aatagtctat 120
tacgccaagt caatccgaag ttcgattgag gatgctaag cattacgaac gagctagacc 180
gtaaaggcca gcagcgctca gaaaccaa acctcttcaa tcattaagaa agaggagggt 240
tgaagtattc atgacactca aacaggcatg ctccacgga taccatggag cgcaagggtc 300
gttcaaagat tcgatgattc actgaattct gcaattcaca ttacttatcg catttcgctg 360
cgttcttcat cgatgcgaga gccaagagat ccgtgttga aagttttatt ttgttataa 420
atttaataca ttcatagact ttgtgtttat aagtgaatag gagttcgctc tcttgcgaga 480
gttactatcc caaacaagt cagagggtta gaaagtgaga gttcggactc caagttaagt 540
tggacgtcct atattcacta atgaccttc cgcaggcccc ctaacgggaa ggagggttta 600
ttattgaaaa atggaaa 617

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<210> SEQ ID NO 17
<211> LENGTH: 1014
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP1/27F, reverse sequence

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<400> SEQUENCE: 17

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cgaggccggg agtgtgatag agaagcttgc ttctcttgat acgttgagcg gcggacgggt	60
gagtaacacg taggcaacct gccctcaagt ttgggacaac taccggaaac ggtagcta	120
accgaataat tgttttcttc gctgaagga aactggaaag acggagcaat ctgtcacttg	180
gggatgggcc tgcggcgcat tagctagttg gtggggtaac ggctcaccaa ggcgacgatg	240
cgtagccgac ctgagagggt gatcgccac actgggactg agacacggcc cagactccta	300
cgggaggcag cagtagggaa tcttcgcaa tgggcgaaag cctgacggag caatgccgcg	360
tgagtgatga aggttttcgg atcgtaaagc tctgttgcca gggaagaacg cttgggagag	420
taactgctct caaggtgacg gtacctgaga agaaagcccc ggctaactac gtgccagcag	480
ccgcggtaat acgtaggggg caagcgttgt ccggaattat tgggcgtaaa gcgcgcgcag	540
gcggtcattt aagtctgggt tttaatcccg gggtcaacc ccggatcgca ctggaaactg	600
ggtgacttga gtgcagaaga ggagagtga attccacgtg tagcggtgaa atgcgtatat	660
atgtggagga acaccagtgg cgaagcgac tctctgggct gtaactgacg cttaaagcgcg	720
taagcgtggg gaacaaaaac gattaatac cctgggttcc cacaccgtaa acgatgaatg	780
ctacgtgtca aagggttctc ccccccttgt gtgcgaaatt aaacacttta cactccccgc	840
ggggagtaga gctccccac ggggatacca ccgagatttg aggggcccc gccccgtttt	900
gaaacttgtt gattagtaag acgccgcga aaaatttata ggggggcagc agctaacaag	960
gcggtggggg tttcttcttc gccccgaaac aagaaaaaat gtcaccctcg aatt	1014

<210> SEQ ID NO 18

<211> LENGTH: 1063

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer PP1/R14192 forward sequence

<400> SEQUENCE: 18

ttgttcggcg gctggctcct tgcggttacc ccaccgactt cgggtgttat aaactctcgt	60
ggtgtgacgg gcggtgtgta caagaccgga gaacgtatcc accgcggcat gctgatccgc	120
gattactagc aattccgact tcatgcaggc gagttgcagc ctgcaatccg aactgagacc	180
ggctttgttg ggattggctc catctcgca tttcgagcc cgttgtaacc gccattgtag	240
tacgtgtgta gccaggtca taaggggcat gatgattga cgtcatcccc accttcctcc	300
ggtttgtcac cggcagtcct tctagagtgc ccaccggaag tgctggcaac taaatataag	360
ggttgcgctc gttgcgggac ttaacccaac atctcacgac acgagctgac gacaaccatg	420
caccacctgt cttgactgtt ccgaagaaaa ggtacatctc tgcaccggtc attcggatgt	480
caagacctgg taagggtctt cgcgttgctt cgaattaaac cacatactcc actgcttggtg	540
cgggtccccg tcaattcctt tgagtctcag tcttgcgacc gtactcccca ggcggagtgc	600
ttaatgtgtt aacttcggca ccaagggat cgaaacccct aacacctagc actcatcggt	660
tacggcgtgg actaccaggg tatctaatac tgtttgctcc ccacgcttcc gcgcctcagc	720
gtcagttaca gccagagag tcgccttcgc cactgggtgtt cctccacata tctacgcatt	780
tcaccgctac acgtggaatt ccactctcct cttctgcaat caagtcaccc agtttccagt	840
gcgatccggg gttgagcccc gggattaaac accagactta aatgaccgcc tgcgcgcgct	900
ttacgcccc aatttcggga caacgcttgc cccctacgta ttaccgggcg tgctggcagc	960

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tatttagccg gggctttctt ctcaggtacc gtcacottga aagcagttac tctcccaagc 1020

gttcttcctt ggcaacagag ctttacgata cgaaaacctt cac 1063

<210> SEQ ID NO 19

<211> LENGTH: 842

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer PP2/M1387R, reverse sequence

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (793)..(793)

<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 19

aggaaaaatg cctaataccc gattttatct ctttccgact gcatgcctgc taggcgaacc 60

gtgcaataac gcgtagacac ccggcctatt gcgagggctc catctcgaga tgatcgcttc 120

ccctgtaggg ccattgttgt acgagtgtaa ccaggtcag ggaggggacg atgatttgac 180

gacctccccg cttctctcgg gtttgcacc ggcactctat ctgactgcc caccgacgt 240

gctggccact aagtataagg gttgcgctcc ttacgcgact taaccacaca tctcacgaaa 300

cttgcctgact agggcatgcc ccacctgtct ctaccgtccc gaagaaaaaa ggcatccctg 360

ccccgatctc tgttatgtca agacaaggca agggctctct cgtggctctc aaagtgaccg 420

taactgacca agaaagcgtg gggtagcccg cgctccttt gaggttaatt cttacggcgc 480

aagtcacccc ccggaagtgt tgggtgtgtt gccttcgcca gcggtggtt aaatatggcg 540

tttaaccccc gggctcacc ccatggggga tgaaaactgg gtgacttgag ggtagatgac 600

gagagtgtaa ttctcgtgg agcgttgaaa tgcgtagata tgtctacgaa caccagggtc 660

gacgcgatct atgggctgta actgacgctg aggagcaca ctcttggtga tctgacagga 720

ttaaatcccc tggtagtcca atccgtagat gatgaccgct gactgtagg ggtcttaaat 780

gactgggtgg cgnaggtata cattaagcac atacagtagg ggaatacgcg tcgccagaa 840

tt 842

<210> SEQ ID NO 20

<211> LENGTH: 951

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer PP1/27F, reverse sequence

<400> SEQUENCE: 20

ggggaaatgg gcgctgcta aacatgcagt cgaacgggtga agcccagctt gctgggtgga 60

tcagtggcga acgggtgagt aacacgtgag caacctgccc ctgactctgg gataagcgct 120

ggaaacggcg tctaatactg gatacgagta gcgaccgcat ggctcagttac tggaaagatt 180

tattgggttg ggatgggctc gcggcctatc agcttggttg tgaggtaatg gctcaccaag 240

gcgtcgacgg gtagccggcc tgagaggggt accggccaca ctgggactga gacacggccc 300

agactcctac gggaggcagc agtggggaat attgcacaat gggcgcaagc ctgatgcagc 360

aacgcgcgct gagggatgac ggccttcggg ttgtaaacct cttttagcag ggaagaagcg 420

aaagtgcagg tacctgcaga aaaagcgccg gctaactacg tgccagcagc cgcgtaata 480

cgtagggcgc aagcgttatc cggaattatt gggcgtaaa agctcgtagg cggtttgtcg 540

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cgtctgctgt gaaatccgga ggctcaacct ccggcctgca gtgggtacgg gcagactaga      600
gtgcggtagg ggagattgga attcctgggtg tagcgggtgga atgcgcagat atcaggagga      660
acaccgatgg cgaaggcaga tctctggggcc gtaactgacg cttaggagcg aaaggggtggg      720
gagcaaacag gcttagatag cctggtagtc caccocgtaa acgttgggaa ctagtgtgtg      780
ggtccattcc acggattccg tgacgcagct tacgcattaa gttccccgcc tggggagtac      840
ggccgcaagg ctaaaactca aaggaattga ctgggacccg cacaagcggc gaaacatgcg      900
gattaattcc atgcaactcc gaagaacctt accagggtct gaacatatac a                951

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<210> SEQ ID NO 21
<211> LENGTH: 978
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP1/R14192, forward sequence

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<400> SEQUENCE: 21

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ggaatttcga tccaccatat cgacggctcc ctccacaagg gttaggccac cggcttcagg      60
tgttaccgac tttcatgact tgacgggctg tgtgtacaag acccggaac gtattcaccg      120
cagcgttgct gatctgcgat tactagcgac tccgacttca tgaggctcgag ttgcagacct      180
caatccgaac tgggacccgc tttttgggat tcgctccacc tcgcggtatt gcagcccttt      240
gtaccggcca ttgtagcatg cgtgaagccc aagacataag gggcatgatg atttgacgtc      300
atccccacct tcctccgagt tgaccccggc agtatcccat gagttccac cattacgtgc      360
tggcaacata gaacgagggt tgcgctcgtt gcgggactta acccaacatc tcacgacacg      420
agctgacgac aacctatcac cacctgttta cgagtgtcca aagagttgac catttctggc      480
ccgttctcgt atatgtcaag ccttggttaag gttctctcgc ttgcatcgaa ttaatccgca      540
tgctccgccc cttgtgcggg tccccgtaaa ttcccttgag ttttagcctt gcggccgtac      600
tccccaggcg gggaaactaa tgcgttagct gcgtcacgga atccgtggaa tggacccac      660
aactagtacc caacgtttac ggggtggact accagggtat ctaagcctgt ttgctcccca      720
ccctttcgct cctcagcgtc agttacggcc cagagatctg ccttcgcat cggtgttctc      780
cctgatattc gcgcattcca ccgctacacc aggaattcca atctccccta ccgcactcta      840
gtctgccccg acccactgca ggccggaggt tgagcctccg gatttcacag cagacgcgac      900
aaaccgccta cgagctcttt acgcccataa attccggaaa acgcttggcg ccctacgtat      960
taccgcggtc gctggccc                978

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<210> SEQ ID NO 22
<211> LENGTH: 629
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP1/63F, reverse sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2)..(2)

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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (18)..(18)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (34)..(34)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 22

gnccgacnag ggttggnntg actggctatt ccangggatt cctattcttg aggtcgagtt      60
agtaaacactc aggtatcctt gctattagct tttatgacta ttggaaacct cgcggaattg      120
cagactatta tttattgcaa tagtatttgg tgggaaaccg aacaatactg ggcaagatga      180
tgtgttgtea tcgtcatttt catccgacat ctccccggcc gtctatcaag agttaccacc      240
attactggct ggcaaaatag aacgagagtt gcgctcgttg cgggacttaa cacaacagct      300
cacgacacga gctgacgaca accatgcgcc acctgtttaa aaatggccct agagttgaaa      360
atttctggte cgttctcata aatgtaaagg cctggtatgt tcttaacgtt gcgctaaata      420
atccccatgc tccgaggatt gtgcagggcc ccgtcgtttt ccttaatttt tagccttgag      480
gacgtactac ccagggtggg aactttgtgg ggtaactggg acgccgaacc ccggaatgga      540
cgcccaacgg catccaaagt ttgtagggga aattcaagtc atctaaaccg gttgatecccc      600
acaattctga ctaaccgatg gcaaggcaa                                     629

<210> SEQ ID NO 23
<211> LENGTH: 1307
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer PP1/M1387R, forward sequence

<400> SEQUENCE: 23

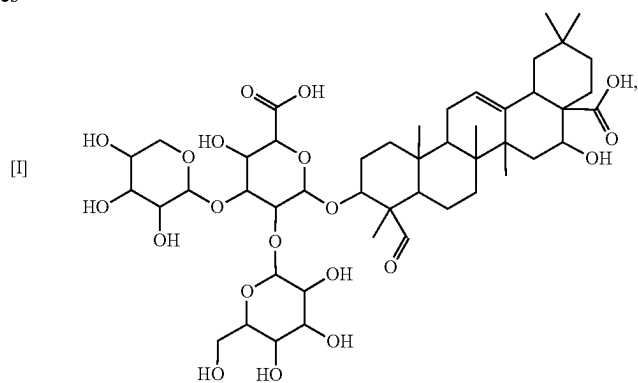
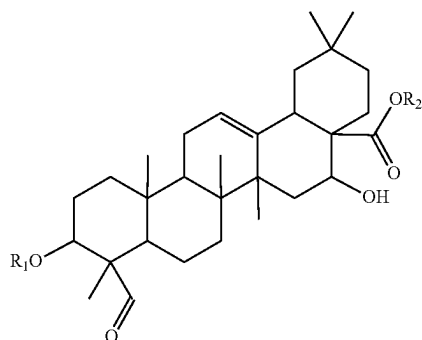
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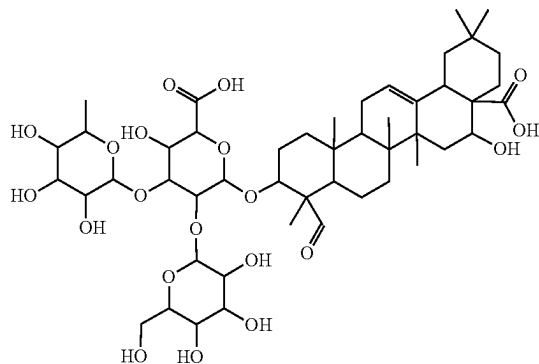
1. Method for preparing of an extract of 3-O-monodesmosidic saponins having formula (I)



quillaic acid 3-O-{rhamnopyranosyl-(1+3)-[galactopyranosyl-(1+2)]-glucopyranosiduronic acid}, compound I,

wherein R_1 is {xylopyranosyl-(1+3)-glucopyranosiduronic acid} or a salt thereof, {xylopyranosyl-(1+3)-glucopyranosiduronate salt}; {rhamnopyranosyl-(1+3)-glucopyranosiduronic acid} or a salt thereof, {rhamnopyranosyl-(1+3)-glucopyranosiduronate salt}; or {galactopyranosyl-(1+2)-glucopyranosiduronic acid} or a salt thereof, {galactopyranosyl-(1+2)-glucopyranosiduronate salt}; or {glucopyranosiduronic acid} or a salt thereof, {glucopyranosiduronate salt}, and R_2 is hydrogen or a cation specie, substantially free of bidesmosidic saponins, comprising:

- obtaining a partially hydrolyzed extract from a plant biomass extract containing glycosides of quillaic acid and obtaining a crude hydrolysate,
- recovering from the crude hydrolysate obtained in step a), an extract of monodesmosidic saponins comprising Compounds I and II,



quillaic acid 3-O-{galactopyranosyl-(1+2)-glucopyranosiduronic acid}, compound II;

- culturing such extract obtained in step b) into a liquid fermentation culture of a microbial consortium (HNR2), wherein such microbial consortium (HNR2) comprising microorganisms selected from *Rhodotorula*

sp having accession number RGM 3031 (colony P) dated Jul. 22, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Microbacterium* sp having accession number RGM 3021 (colony Y) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Mucor* sp having accession number RGM 2984 (colony C) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Paenibacillus* sp and having accession number RGM 3032 (colony W) dated Sep. 3, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM), and

d) removing such microbial consortium (HNR2) from such fermentation broth obtained in step c) and recovering an *Quillaja* extract substantially free of microorganisms and bidesmosidic saponins.

2. The method of claim 1 wherein such plant biomass extract containing glycosides of quillaic acid is selected from an extract of bark, pruned branches and leaves of *Quillaja* plant.

3. The method of claim 2 wherein such *Quillaja* plant is selected from *Quillaja saponaria* Molina, *Quillaja brasiliensis* and *Saponaria vaccaria*.

4. The method of claim 2 wherein such extract of *Quillaja* plant is an extract of *Quillaja saponaria* Molina.

5. The method of claim 1 further comprising supplementing the fermentation broth with oligosaccharides of xylose at step c).

6. The method of claim 1 further comprising isolating such 3-O-monodesmosidic saponins having formula (I) from such extract substantially free of microorganisms and bidesmosidic saponins as resulting from step d), by solvent extraction followed by solvent evaporation at reduced pressure, or by chromatography followed by solvent removal at reduced pressure and lyophilization.

7. The method of claim 1 wherein such partially hydrolyzed extract of step a) is obtained from a plant biomass extract containing glycosides of quillaic acid selected of an aqueous *Quillaja* extract.

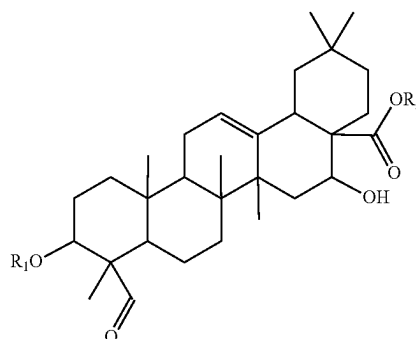
8. The method of claim 7 wherein such partially hydrolyzed extract of step a) is obtained by KOH hydrolysis from an aqueous *Quillaja* extract.

9. The method of claim 1 wherein such crude hydrolysate is obtained from submitting to aqueous ethanol, such partially hydrolyzed extract of *Quillaja* plant.

10. The method of claim 1 further comprising prior to step b), fractionating such crude hydrolysate to obtain a crude hydrolysate enriched in Compound I and Compound II.

11. The method of claim 10 wherein such fractionating is performed by chromatography on a hydrophobic resin.

12. An extract of 3-O-monodesmosidic saponins, substantially free of bidesmosidic saponins, stable at physiological pH and reduced hemolytic effect, having formula (I)



[I]

wherein R_1 is {xylopyranosyl-(1+3)-glucopyranosiduronic acid} or a salt thereof, {xylopyranosyl-(1+3)-glucopyranosiduronate salt}; {rhamnopyranosyl-(1+3)-glucopyranosiduronic acid} or a salt thereof, {rhamnopyranosyl-(1+3)-glucopyranosiduronate salt}; or {galactopyranosyl-(1+2)-glucopyranosiduronic acid} or a salt thereof, {galactopyranosyl-(1+2)-glucopyranosiduronate salt}; or {glucopyranosiduronic acid} or a salt thereof, {glucopyranosiduronate salt}, and R_2 is hydrogen or a cation specie.

13. A microbial consortium to culturing *Quillaja* extracts comprising microorganisms selected from *Rhodotorula* sp having accession number RGM 3031 (colony P) dated Jul. 22, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Microbacterium* sp having accession number RGM 3021 (colony Y) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Mucor* sp having accession number RGM 2984 (colony C) dated Aug. 7, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM); *Paenibacillus* sp and having accession number RGM 3032 (colony W) dated Sep. 3, 2020 of the Chilean Collection of Microbial Genetic Resources (CChRGM).

14. The microbial consortium of claim 13 further comprising one or more of *Paenibacillus amylolyticus*, *Paenibacillus tundra*, *Paenibacillus tylopili*, *Microbacterium hydrocarbonoxydans*, *Microbacterium luteolum*, *Microbacterium phyllosphaerae*, *Rhodotorula mucilaginosa*, *Mucor rcemosus*, *Cylindrocarpon* spp., *Aspergillus niger* and *Aspergillus parasiticus*.

15. A method for treating cancer comprising contacting cancer cells with an extract as defined in claim 12.

16. The method of claim 15 wherein such cancer cells are gastric cancer cells.

17. The method of claim 16 wherein such gastric cancer cells are adenoma cells and carcinoma cells.

18. A method for treating oral infections produced by oral pathogens comprising contacting cancer cells with an extract as defined in claim 12.

19. The method of claim 18 wherein such oral pathogens is *Streptococcus mutants*.

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